What Investment Data Implies about the AI Transition*

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Abstract

Five U.S. technology firms account for the majority of AI infrastructure investment. They spent $413 billion on capital expenditure in 2025 and are forecast to spend roughly double that in 2026. These firms risk bankruptcy unless expected profits grow commensurately. We embed this observation in a two-sector open-economy model with rare productivity booms. We calibrate the boom size to match the capital stock implied by investment projected through 2027, implying that a boom raises AI-sector productivity by a factor of roughly 3.5. We then calibrate a two-year window of a 50% annual probability of an increase of the same magnitude, generating a range of scenarios consistent with the wide variety of industry forecasts, along with an elevated permanent probability tied to the valuation of the aggregate market. The implied additional cumulative GDP growth, once capital has fully adjusted after this window, ranges from 8 to 133 percentage points, with AI shares of the economy ranging from 10% to 58%. Long-term annual growth is in expectation approximately 7% but with substantial risk. With risk aversion of 3, and an elasticity of intertemporal substitution equal to 1, the risk-free rate increases by approximately a third of a percentage point, and the equity premium rises by approximately 3.5 percentage points.

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[1]An interactive companion to this paper is available at https://jwachter3.github.io/research/ai-transition/. We thank Patrick Adams, Sylvain Catherine, Gary Gensler, Amy Huber, Alex Imas, Ioana Marinescu, Max Miller, Sean Myers, Stijn Van Nieuwerburgh, Fernando Stein, and seminar participants at Wharton for helpful comments. We thank Frank Ma for valuable research assistance. The information, views, and opinions expressed herein are solely those of the authors and do not necessarily represent the views of Point72 or its affiliates. Point72 and its affiliates are not responsible for, and did not verify for accuracy, any of the information contained herein.

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1 Introduction

How much will artificial intelligence contribute to economic growth? Existing answers span an enormous range. Acemoglu (2024) applies a task-based framework and concludes that AI will raise total factor productivity by approximately 0.7 percentage points over ten years. At the other extreme, Amodei (2024, 2026) envisions industry-wide infrastructure spending reaching "multiple trillions a year by 2028 or 2029," with AI compressing decades of scientific progress into years.

Our contribution is to use investment data, combined with revealed preference, and a model for rare productivity booms, to estimate future GDP growth. Assuming the marginal investment is (conservatively) zero net present value implies a near-term increase in AI-sector productivity of a factor of roughly 3.5—this is a rare boom of the type studied by Tsai and Wachter (2016). We then use industry forecasts to 2030 to argue that firms are considering additional booms—but that these are far from certain. Finally, we calibrate a long-term boom probability to aggregate stock market valuations.

This approach complements the growing literature that focuses on task-based analyses, such as Acemoglu (2024), Babina et al. (2024), and Yotzov et al. (2026). The productivity implications from the current investment numbers are significantly greater than those implied by the task-based analyses, even those using more recent data. A second strand of the literature (Korinek and Suh, 2024, Restrepo, 2025, Trammell and Korinek, 2023) focuses on how wages might respond as human work becomes less necessary. A limiting case is one in which the physical and intelligence sectors are gross complements (Kording and Marinescu, 2025), implying that AI-related growth eventually hits a wall. The extent to which these two sectors are gross complements is itself highly uncertain, and our model captures this possibility through the extreme uncertainty regarding whether further booms arrive or not. Our paper also relates to recent work by Chen (2026), who focuses on the possibility of

the singularity as a rare disaster. Rather than assuming the singularity, we back out the size of the initial boom from investment data of value-maximizing firms using a revealed-preference argument. The singularity is the scenario in which two further booms of this size occur. Finally, Van Nieuwerburgh (2026) examines how firms finance the AI buildout. Our approach differs from these in that we focus on implications for productivity and the share of AI in the economy, assuming that wages and financing frictionlessly adjust.

Section 2 documents the facts. A small number of firms—Amazon, Alphabet, Microsoft, Meta, and Oracle—account for the vast majority of AI infrastructure spending. Their combined capital expenditure was $159 billion in 2022, rose to $242 billion in 2024, and is forecast at $784 billion for 2026. The 2022 and 2024 figures come from the firms' SEC filings. The 2026 figure combines management forecasts for Amazon, Alphabet, and Meta with, for Microsoft and Oracle, capital expenditure in the first part of the year plus an extrapolation for the remainder. Section 2 reports these figures company by company and places them in the context of the aggregate economy.

Section 3 develops a two-sector model with rare productivity booms. The AI sector is exposed to the boom; the non-AI sector is not. The initial boom entails an unanticipated jump in productivity, to which optimizing firms respond with a surge in investment. To justify investment through 2027, the boom size is roughly a multiple of 3.5 of current productivity. This boom size informs the remaining calibration. With no other changes to productivity, this multiple alone implies a permanent increase in the level of GDP of approximately 8% by 2030.

However, industry forecasts, together with historical experience regarding other fundamental technological changes, suggest that the current productivity boom is not the end of the story. The rare boom model of Tsai and Wachter (2016) provides a means of capturing the skewness and optionality inherent in the future trajectory of AI. We model the occurrence of future advances as the outcome of a Bernoulli process with maximal uncer-

tainty. We trace out a range of scenarios over 2028–2030, which we refer to as moderate, transformative, and singularity. The moderate scenario corresponds to the 8% permanent GDP gain mentioned above, already an order of magnitude above task-based estimates. By 2030, the transformative and singularity scenarios raise GDP by approximately 35% and 84%, respectively, relative to the pre-boom economy.

We then use the model, applied to current stock market valuations, to obtain a back-of-the-envelope annual probability of a boom after 2030. This probability is approximately 3 percent per annum. This in turn allows us to compute long-run distributions, in which, under the moderate scenario, growth accumulates to 40 percentage points by 2050 along the expected path, but with a 90th percentile value of 114%, whereas for the singularity, the expected path corresponds to 522% growth, with a 90th percentile of about 1,415%. Not surprisingly, this level of uncertainty has strong implications for the risk-free rate and equity premium. Despite the higher expected growth, the equilibrium risk-free rate barely increases, and the equity premium increases by 3.5 percentage points due to uncertainty. We map out these implications in Section 5. Finally, Section 6 places the AI boom in the context of historical growth episodes. Section 7 concludes.

2 AI capital expenditure

We compile capital expenditure data from large publicly traded firms. Because AI infrastructure investment is highly concentrated, it is possible to measure the sector's investment directly from these firms' regulatory filings and management forecasts.

Table 1 reports annual capital expenditure for the five firms that account for the majority of AI infrastructure spending: Amazon, Alphabet, Microsoft, Meta, and Oracle. All columns are on a calendar-year basis, except that Oracle's figure for each year covers the twelve months from December of the preceding year through November. The 2022–2025 figures are drawn

from each firm's 10-K filings with the SEC and, for Microsoft and Oracle, their 10-Q filings; the 2026 and 2027 figures are forecasts and estimates, as described in the notes to the table.

Table 1: Capital expenditure of major AI infrastructure firms ($ billions)

	2022	2023	2024	2025	2026 (forecast)	2027 (est.)
Amazon	64	53	83	132	200	285
Alphabet	31	32	53	91	185	255
Microsoft	25	35	56	83	169	250
Meta	32	28	39	72	138	195
Oracle	7	7	11	35	92	105
Total	**159**	**155**	**242**	**413**	**784**	**1,090**

Notes: All columns are on a calendar-year basis; Amazon, Alphabet, and Meta report on the calendar year. For Microsoft (fiscal year ending June 30) and Oracle (fiscal year ending May 31), the figure for year T is the figure for the fiscal year ending in year T, minus the first six months of that fiscal year, plus the first six months of the following fiscal year. For Microsoft this is the calendar year; for Oracle, whose quarters end in August, November, February, and May, it is the twelve months from December of year $T-1$ through November of year T. Values for 2022–2025 are cash capital expenditures from the Consolidated Statements of Cash Flows in each firm's Form 10-K and, for Microsoft and Oracle, Form 10-Q. The 2026 figures for Amazon, Alphabet, and Meta are management forecasts (midpoints of disclosed ranges for Alphabet and Meta). The 2026 figure for Microsoft is capital expenditure in January–June 2026 plus an extrapolation for July–December 2026; the 2026 figure for Oracle is capital expenditure in December 2025–August 2026 plus an extrapolation for September–November 2026. The 2027 figures are bottom-up estimates built from multi-year commitments already disclosed in filings—leases "not yet commenced," contracted revenue, and signed multi-year customer agreements. Each total is the sum of the entries shown. Appendix A.1 documents the sources, the calendar-year construction, and the firm-specific measurement conventions for the 2022–2025 figures, and Appendix A.2 the 2026 and 2027 figures.

The values reported in Table 1 suggest that something unusual occurred in 2024. Aggregate capital expenditure was $159 billion in 2022 and $155 billion in 2023, reflecting baseline spending on cloud computing infrastructure and general corporate investment. This baseline encompasses non-AI uses—retail fulfillment, logistics, office infrastructure, and legacy cloud services—alongside any pre-boom AI spending. As discussed in Section 3.4, assigning this full amount to AI spending is conservative. In 2024, aggregate spending rose 56% to $242 billion. The acceleration continued in 2025, with a further 71% increase to $413 billion, and

the 2026 figures imply a further 90% increase to approximately $784 billion.[4] Altogether, capital expenditure will have more than tripled between 2024 and 2026.

The five firms in Table 1 do not span the universe of AI infrastructure capital spend. A sixth hyperscaler (xAI, founded mid-2023, now SpaceX's AI segment) and four data-center-owning "neocloud" providers (CoreWeave, Crusoe, IREN, Lambda) add approximately $17 billion in 2024 and $41 billion in 2025 of capital expenditure not reflected in Table 1. Excluding this additional investment is also conservative. Appendix A.3 documents these firms and their accounting treatment.

Two caveats apply to Table 1. First, the five firms define capital expenditure somewhat differently. The largest difference concerns equipment acquired through leases rather than purchased outright: Meta adds principal payments on its finance leases to the total it discloses, and the remaining four firms do not include leased equipment in their reported lines. Appendix A.1 states each firm's definition and matches each value in Table 1 to the firm's cash-flow statement. Because each firm applies its definition consistently over time, and because the calibration in Section 3.3 depends on the ratio of the capital stock entering 2028 to the pre-boom stock, both built from the same investment series, these differences have little effect on the results. Second, all investment values are nominal. We treat them as real throughout the analysis: cumulative inflation over 2024–2027 is on the order of 8–12%, small relative to the orders of magnitude at stake.

The 2027 estimates in Table 1 are built from multi-year commitments disclosed in filings—data-center leases signed but not yet in effect, contracted revenue that requires new capacity, and signed customer agreements. Microsoft, for example, reported $329 billion in leases "not

[4]The 2026 figures are: Amazon $200B, Alphabet $185B, Microsoft $169B, Meta $138B, and Oracle $92B. Amazon, Alphabet, and Meta have all disclosed quantitative annual guidance in Form 8-K filings; the figures for Alphabet and Meta are midpoints of the disclosed ranges. Microsoft's figure is $66.7B of additions to property and equipment in January–June 2026 plus $102B for July–December 2026, extrapolated from the $35.8B reported for April–June 2026. Oracle's figure is $63.6B of capital expenditures in December 2025–August 2026 plus $28.5B for September–November 2026, the amount reported for June–August 2026. Appendix A.2 reports the source documents and the extrapolations.

yet commenced" as of June 30, 2026, up from $93 billion a year earlier; the filing states that some of these arrangements are subject to contractual conditions. Amazon disclosed commitments from OpenAI and Anthropic for Trainium capacity beginning in 2027. Appendix A.2 documents the construction firm by firm.

Table 2: AI infrastructure investment relative to the U.S. economy

	2022	2024	2025	2026 (forecast)	2027 (est.)
AI infrastructure capex ($B)	159	242	413	784	1,090
U.S. nominal GDP ($T)	26.1	29.4	30.9	∼31.8	∼32.7
U.S. gross private fixed investment ($T)	4.7	5.3	5.6	∼5.7	∼5.9
Share of GDP (%)	0.6	0.8	1.3	2.5	3.3
Share of private fixed investment (%)	3.4	4.6	7.4	13.7	18.5

Notes: GDP and gross private fixed investment for 2022–2025 are from the Bureau of Economic Analysis (NIPA Table 1.1.5, as published in September 2026; FRED series GDPA and FPIA); the 2026–2027 values extend the 2025 figures at 3% nominal annual growth. Shares are computed from the unrounded values. AI infrastructure capex is the five-firm total from Table 1; Appendix A.2 documents the 2026 and 2027 figures.

Table 2 places these figures in the context of the U.S. macroeconomy. The capital expenditure values are consolidated worldwide totals: they include data centers these firms build outside the United States, whereas GDP and gross private fixed investment count only investment located in the United States. To the extent that some of the spending occurs abroad, the ratios overstate the U.S. shares. AI infrastructure investment rose from 3.4% of total U.S. private fixed investment in 2022 to a projected 14% in 2026. As a share of GDP, it rose from 0.6% to 2.5%.

3 A two-sector model with rare booms

This section develops a two-sector model in which the AI sector is exposed to rare productivity booms and the non-AI sector is not. The production framework builds on models

developed by Gourio (2012) and Gomes et al. (2019), who study firms subject to rare disasters. We depart from these models in two ways. First, we replace disasters with booms. Second—and more consequentially for investment—we break the assumption, introduced by Gourio (2012), that the shock moves productivity and the capital stock proportionally. In the disaster setting, the shock destroys both z and K in proportion, leaving the marginal product of capital unchanged; investment does not respond. Here, the boom raises productivity but leaves the physical capital stock intact, so that the marginal product of capital jumps upward and firms face a large gap between current and optimal capital. This asymmetry is the source of the investment surge.

3.1 Two sectors

The economy consists of two sectors: AI (a) and non-AI (n). Each sector $j \in \{a, n\}$ produces output using a Cobb-Douglas technology:

$$Y_{jt} = z_{jt}^{1-\alpha} K_{jt}^{\alpha}, \tag{1}$$

where α is the capital share, z_{jt} is sector-specific productivity, and K_{jt} is the sector's capital stock. Capital in each sector evolves according to

$$K_{j,t+1} = (1 - \delta_j) K_{jt} + I_{jt}, \quad j \in \{a, n\}, \tag{2}$$

where I_{jt} is gross investment and δ_j is the sector-specific depreciation rate. Assuming that each firm begins at its optimal level of capital before seeing the shock, the problem is scale invariant, and it suffices to consider a representative firm's optimization problem.

Productivity in the AI sector is subject to rare booms. In each period, with probability p_t, a boom occurs: AI productivity jumps by a factor e^{ξ}, where $\xi > 0$ represents the log

change in productivity due to the boom. The non-AI sector is unaffected. Formally:

$$z_{a,t+1} = \begin{cases} z_{at}\, e^{\xi} & \text{with probability } p_t, \\ z_{at} & \text{with probability } 1 - p_t. \end{cases} \tag{3}$$

For convenience, we assume neither sector is affected by normal-times volatility or by drift. We allow p_t to vary over time; in the calibration below, the variation takes the form of a structural break. Note that p_t determines the probability of the boom that occurs at time $t+1$. Agents choose investment I_{jt}, and hence capital $K_{j,t+1}$, with knowledge of z_{at} and p_t, which determine the *distribution* of $z_{a,t+1}$, but without knowledge of $z_{a,t+1}$ itself. Thus capital $K_{a,t+1}$ is known at time t.

3.2 Optimal investment

Firms in the AI sector maximize the present discounted value of output net of investment, assuming discount rate r_a; the non-AI sector's problem is analogous, with discount rate r_n.

$$V_{at} = \max_{\{I_{as}\}_{s \geq t}} \mathbb{E}_t \sum_{s=t}^{\infty} \frac{1}{(1+r_a)^{s-t}} \left[z_{as}^{1-\alpha} K_{as}^{\alpha} - I_{as} \right], \tag{4}$$

subject to $K_{a,s+1} = (1-\delta_a)K_{as} + I_{as}$ and the productivity process (3).[5] Sectors face different exogenous required rates of return, r_a and r_n, reflecting differences in risk. Following an

[5]The production function $Y_j = z_j^{1-\alpha} K_j^{\alpha}$ can be interpreted as Cobb-Douglas in capital and labor, $Y_j = A_j K_j^{\alpha} L_j^{1-\alpha}$, where A_j is total factor productivity, the sector's labor supply is fixed at $\bar{L}_j$, and $z_j = A_j^{1/(1-\alpha)} \bar{L}_j$. The problem above can then be viewed in two ways. The first, which we adopt in the text, is an optimization problem with no labor: z_{at}, and with it $\bar{L}_a$, is fixed. The second is an optimization problem at a fixed wage. A firm with capital K takes the wage w as given and decides it can hire as much labor as it wants at that wage; it does not internalize the externality it imposes on the sector through the wage. Writing $F(K,L) = A_a K^{\alpha} L^{1-\alpha}$, its profit is

$$\pi(K; w) = \max_{L} \left[F(K,L) - wL \right] = \phi(w)\, K, \qquad \phi(w) = \alpha A_a^{1/\alpha} \left(\frac{1-\alpha}{w} \right)^{(1-\alpha)/\alpha},$$

"open economy" interpretation, we treat these as exogenous—determined, for instance, by the required return of global investors willing to supply capital elastically at price r_j. In Section 5, we derive implications of rare booms for equilibrium rates of return.

For the firm's problem to have a finite solution, the expected gross growth factor of productivity must be below the gross discount rate:

$$1 + r_a > (1 - p) + p\, e^{\xi}. \tag{5}$$

Given that p is subject to unanticipated structural breaks, (5) may be violated for a value of p that obtains for a finite number of periods. We verify that this condition holds at the calibrated parameters below, when p is at its long-run value.

The first-order condition with respect to $K_{a,t+1}$, or equivalently, I_{at}, equates the marginal cost of a unit of capital (one, since there are no adjustment costs) to the discounted expected marginal product plus the continuation value of the undepreciated unit:

$$1 = \frac{1}{1 + r_a}\, \mathbb{E}_t \left[\alpha \left(\frac{z_{a,t+1}}{K_{a,t+1}} \right)^{1-\alpha} + (1 - \delta_a) \right],$$

which is linear in the firm's capital, with optimal labor $L^*(K, w) = K\left((1-\alpha)A_a/w\right)^{1/\alpha}$. Differentiating with respect to K at the fixed wage, with L^* responding to K,

$$\frac{\partial \pi}{\partial K} = F_K + (F_L - w)\, \frac{\partial L^*}{\partial K} = F_K,$$

where the second term is zero by the first-order condition for labor (the envelope theorem), and there is no term in $\partial w/\partial K$ because the firm takes the wage as given. Freely adjusting labor thus leads to a first-order condition for investment, $\mathbb{E}_t[\phi(w_{a,t+1})] = r_a + \delta_a$, in which the firm's capital does not appear. Capital comes back in through the sector-wide wage. Summing labor demand over firms, with sector capital K_a and labor supply $\bar{L}_a$, the labor market clears at $w_a = (1-\alpha)A_a(K_a/\bar{L}_a)^{\alpha}$, so that $\phi(w_a) = \alpha(z_a/K_a)^{1-\alpha}$, the marginal product of sector capital, giving the same first-order condition as the original problem, which we derive below. At this wage the wage bill $w_a\bar{L}_a$ equals $(1-\alpha)Y_a$ and earnings net of wages equal αY_a. Investment is therefore the same whether earnings are measured gross or net of wages, while the level of V_{at} is lower by the present value of the wage bill.

which simplifies to

$$\mathbb{E}_t\left[\alpha\left(\frac{z_{a,t+1}}{K_{a,t+1}}\right)^{1-\alpha}\right] = r_a + \delta_a. \tag{6}$$

The expected marginal product of capital equals the user cost: the required return plus depreciation. Because $K_{a,t+1}$ is determined at t, the expectation is over the boom. Evaluating:

$$\alpha\left(\frac{z_{at}}{K_{a,t+1}}\right)^{1-\alpha} \underbrace{\left[(1-p_t) + p_t\, e^{(1-\alpha)\xi}\right]}_{\equiv\, \Omega(p_t)} = r_a + \delta_a, \tag{7}$$

where we define the boom option value:

$$\Omega(p) \equiv (1-p) + p\, e^{(1-\alpha)\xi}. \tag{8}$$

Note that $\Omega(p) \geq 1$, with equality when $p = 0$. A higher boom probability raises the effective expected productivity of capital, making investment more attractive.

Solving (7) for optimal capital:

$$K^*_{a,t+1} = z_{at}\left(\frac{\alpha\,\Omega(p_t)}{r_a + \delta_a}\right)^{1/(1-\alpha)}. \tag{9}$$

The optimal capital stock is proportional to current productivity z_{at} and increasing in the boom option value $\Omega(p_t)$. The non-AI sector satisfies the analogous condition with $\Omega = 1$ (no boom exposure):

$$K^*_{n,t+1} = z_{nt}\left(\frac{\alpha}{r_n + \delta_n}\right)^{1/(1-\alpha)}. \tag{10}$$

Proposition 3.1 (Capital ratio)**.** *Suppose a boom occurs at time t, so that $z_{at} = z_{a,t-1}\, e^{\xi}$, and that the boom probability shifts from p_{t-1} to p_t. Then*

$$\frac{K^*_{a,t+1}}{K^*_{at}} = e^{\xi}\left(\frac{\Omega(p_t)}{\Omega(p_{t-1})}\right)^{1/(1-\alpha)}. \tag{11}$$

Proof. From (9), $K^*_{a,t+1} \propto z_{at} \cdot \Omega(p_t)^{1/(1-\alpha)}$. The boom multiplies z_a by e^{ξ}, and the structural break shifts p from p_{t-1} to p_t. The $r_a + \delta_a$ term cancels because it appears in both numerator and denominator. □

Under the open-economy assumption, the capital ratio depends only on the boom size ξ and the shift in p, not on r_a. This means we can identify ξ from investment data without taking a stand on the required return.

How a boom affects sectoral output. Because $K_{a,t+1}$ is chosen at time t, capital is predetermined when the boom is realized at $t+1$. The output response therefore occurs in two stages.

In the *impact period* $(t+1)$, capital is fixed at $K^*_{a,t+1}$ but productivity has jumped to $z_{at}\, e^{\xi}$. Output is

$$Y_{a,t+1} = (z_{at}\, e^{\xi})^{1-\alpha}\, (K^*_{a,t+1})^{\alpha} = e^{(1-\alpha)\xi}\, z_{at}^{1-\alpha}\, (K^*_{a,t+1})^{\alpha}.$$

Only the fraction $(1-\alpha)$ of the productivity gain is reflected in output because the capital share α of the production function operates on the predetermined capital stock. At $\alpha = 1/3$ and $\xi \approx 1.25$, $e^{(1-\alpha)\xi} = 2.30$: output rises 130%.

Firms set $K^*_{a,t+2} = e^{\xi}\, K^*_{a,t+1}$, the optimal level of capital given the boom observed at time $t+1$ and an unchanged boom probability. If no further boom occurs, output at $t+2$ becomes

$$Y_{a,t+2} = (z_{at}\, e^{\xi})^{1-\alpha}\, (e^{\xi}\, K^*_{a,t+1})^{\alpha} = e^{\xi}\, z_{at}^{1-\alpha}\, (K^*_{a,t+1})^{\alpha}.$$

The full factor e^{ξ} is now realized: with $\xi \approx 1.25$, $e^{\xi} = 3.48$. The remaining factor of $e^{\alpha\xi} = 1.52$ comes from capital adjustment.

Thus α governs the split between the immediate and delayed output responses. A higher capital share means more of the productivity gain is deferred until capital adjusts. Output

jumps discretely at boom dates (the $(1-\alpha)\xi$ component) and then rises again one period later as capital adjusts (the $\alpha\xi$ component).

3.3 Calibration

To operationalize the above model, we require a value for α and a value for the AI-capital depreciation rate δ_a. To derive GDP implications in Section 4, we require further assumptions: the depreciation rate for non-AI capital, and the rates of return in the two sectors, which we take as fixed, following an open-economy interpretation.

Table 3 summarizes the assumed parameter values. The capital share is set to the standard value $\alpha = 1/3$. The AI-capital depreciation rate, $\delta_a = 0.25$, reflects the shorter economic life of computing equipment versus general infrastructure; this value of δ_a implies that roughly 32% of the capital stock remains after 4 years. Appendix B discusses this calibration and robustness to a lower rate of depreciation. The non-AI depreciation rate $\delta_n = 0.07$ is the standard value in macroeconomic calibrations (Cooley and Prescott, 1995). Finally, we assume a required AI-sector return r_a of 15%, similar to industry estimates (see, e.g., Vanguard Group 2025), and a required non-AI return r_n of 5%.

Table 3: Parameter values

Parameter	Value	Description
α	1/3	Capital share
δ_a	0.25	AI-capital depreciation rate
δ_n	0.07	Non-AI-capital depreciation rate
r_a	0.15	Required real return on AI-sector capital
r_n	0.05	Required real return on non-AI capital

Notes: Depreciation rates and required returns are annual.

Before the boom, we assume both sectors are in steady state, so that investment exactly replaces depreciation: $I_j = \delta_j K_j$. The pre-boom AI capital stock is then $K_{a0} = I_{a,\text{pre}}/\delta_a =$

159/0.25 = \$636B, where \$159B is the combined capital expenditure of the five largest AI firms in 2022 (Table 1).[6] For the non-AI sector, aggregate U.S. gross private fixed investment was approximately \$4.7 trillion in 2022 (Table 2), giving $K_{n0} = (4{,}700 - 159)/0.07 \approx$ \$65,000B. The AI sector's share of the total capital stock is therefore only 1.0%, despite accounting for 3.4% of investment—the wedge arises entirely from the difference in depreciation rates.

2024–2027. Analogously to the rare-disasters approach of Barro (2006), we simplify the calibration by collapsing a multi-period event into an instant of time in our initial model. While the model will not succeed in matching dynamics, it will transparently convey orders of magnitude.

In this frictionless model, capital adjusts immediately to its optimal level each period. We assume the boom probability is unchanged at the time of the initial boom and through the 2027 investment decision, $p_t = p_{t-1}$, so that the Ω terms in (11) cancel and Proposition 3.1 gives $K^*/K_{a0} = e^{\xi}$: the optimal capital stock scales linearly with z.

Table 1 shows investment unchanged between 2022 and 2023, and a rapid increase thereafter. We interpret this investment as the firms' response to a productivity increase that occurred prior to 2024. As (9) shows, an increase in productivity by the factor e^{ξ} implies a gap between actual and desired capital. The frictionless model closes this gap in one year; in the data, investment rises in each year through 2027. We therefore calibrate ξ to the level of the capital stock rather than to the timing of investment: we assume that capital reaches its target by the end of 2027, so that the capital stock entering 2028 equals $K^* = K_{a0}e^{\xi}$.

The capital stock entering 2028 follows from the investment in Table 1 and the law of motion (2). Investment in 2022 replaces depreciation, so that the capital stock entering 2023

[6] We use 2022 because it is the last full year before the structural break: combined capital expenditure was essentially flat between 2022 (\$159B) and 2023 (\$155B), indicating that the boom had not yet affected investment decisions.

is K_{a0}, and investment in each year is in place at the start of the following year. Then

$$K_{a,2028} = (1-\delta_a)^5 K_{a0} + \sum_{t=2023}^{2027} (1-\delta_a)^{2027-t} I_{at}, \tag{12}$$

and our revealed-preference estimate of ξ equals

$$e^{\hat{\xi}} = \frac{K_{a,2028}}{K_{a0}}. \tag{13}$$

With the totals in Table 1, $K_{a,2028}$ = \$2,212B; Appendix A.1 gives the capital stock entering each year. Substituting in,

$$e^{\hat{\xi}} = \frac{2{,}212}{636} \approx 3.48, \qquad \hat{\xi} \approx 1.25.$$

This is, approximately, the cumulative increase in productivity required to justify this level of investment, assuming unchanged discount rates and zero NPV at the margin.[7] Were the boom smaller than this, marginal investment would be NPV-negative. We report $e^{\hat{\xi}}$ and $\hat{\xi}$ to two decimals; the calculations that follow use the unrounded values.

2028–2030. To capture the range of possibilities within the model, we consider that over a short period of time, p is at the level corresponding to maximal uncertainty, that is, $p = 0.5$.[8]

[7] If firms restrain their investment response to preserve monopoly power, a possibility envisioned by Korinek and Suh (2024), the observed capital expansion would imply a larger productivity gain. Moreover, we argue below that the discount rate, if anything, increased during this period.

[8] Existing forecasts of transformative AI exhibit very large uncertainty over the relevant horizon. Metaculus community forecasts place the median date for artificial general intelligence at approximately 2028, with a 25th–75th percentile range from late 2026 to 2031 (Metaculus question 5121, accessed March 2026); this median has fallen from approximately 2060 to 2028 over the past four years. Grace et al. (2024) survey 2,778 AI researchers and find a 50% probability of machines outperforming humans at every task by 2047—13 years earlier than in the same survey conducted in 2022—with a 10% probability by 2027. The interquartile range across individual responses spans decades. Section 3.4 compares the model paths to specific industry forecasts.

We assume:

$$p_t = \begin{cases} p_{\text{pre}} \approx 0 & \text{for } t < 2028, \\ p_{\max} & \text{for } t \in \{2028, 2029\}, \\ p_* & \text{for } t > 2029. \end{cases} \tag{14}$$

Note that the elevated boom probability refers to the productivity boom of the *following* year. Thus $p_{2028} = p_{\max}$ means there is a 50% probability of a boom in 2029, and $p_{2029} = p_{\max}$ means there is a 50% probability of a boom in 2030. Because the elevated-probability window begins in 2028, after the calibration period, 2024–2027 investment is driven entirely by the initial boom. The calibration of ξ is therefore independent of $p_{\max}$ and p_*.

Several comments are in order. First, it is admittedly unrealistic to have a structural break (14) in 2028 unanticipated through 2027. This assumption allows us to transparently calibrate ξ to investment through 2027. If a shift in the probability were anticipated earlier, then this would also drive some of the investments. The effect would be a somewhat lower value of ξ. Second, the assumption $p_{\text{pre}} \approx 0$ is for convenience; it cannot be exactly zero because then a boom could not have occurred. It is realistic to believe it is low, as very few foresaw the growth potential of AI in 2024 (and even those who did revised their views upward). Note that it does not affect our prior calculations because we do not take a stance that this value of p changed between 2024 and 2027. Finally, the order-of-magnitude implications of our theory do not depend on a particular choice of interval. The advantage of picking this one is that there are forecasts, however disputed, and that some finite interval allows us to have a probability that captures the full measure of the uncertainty without violating transversality conditions, which would lead to infinite values.

While we separate out the initial investment period from the ongoing uncertainty for convenience, at the core of our calculation is the idea that there is a boom that is followed by an increase in probability. This idea follows from Bayesian updating (Wachter and Zhu,

2025), and is also consistent with recency playing a role in expectations formation (Wachter and Kahana, 2024).

Assuming a probability of $p_{\max} = 0.50$ gives maximal uncertainty.[9] With two independent draws at $p_{\max} = 0.50$, there are approximately three outcomes,[10] shown in Table 4. The moderate scenario is that the initial boom was a one-time event and no further breakthroughs occur. The transformative scenario is that exactly one additional boom arrives during the window. The singularity scenario is that two additional booms occur, compounding the productivity gains.

Table 4: Scenarios over the 2028–2030 window

Scenario	Additional booms	Probability
Moderate	0	$(1 - 0.5)^2 = 0.25$
Transformative	1	$2 \times 0.5 \times 0.5 = 0.50$
Singularity	2	$0.5^2 = 0.25$

Notes: Scenarios are defined by the number of additional booms realized during the elevated-probability window (14): two independent draws at $p_{\max} = 0.50$, governing booms in 2029 and 2030. The transformative scenario aggregates the two single-boom paths.

In the moderate scenario, the productivity gain from the initial boom is all there is. In the transformative scenario, one additional boom raises productivity by another factor of $e^{\xi} \approx 3.5$. In the singularity scenario, two additional booms raise it by a factor of $e^{2\xi} \approx 12.1$ beyond the initial level. During 2029–2030, additional booms may or may not be realized. From 2031 onward, booms arrive with probability p_*. Absent further booms, capital converges to its post-window steady state.

[9] A Bernoulli random variable has maximal variance at $p = 0.50$. This assumption imposes the least informative prior on whether an additional boom occurs in any given year of the window.

[10] Approximately, because the model's behavior is not invariant to whether a single boom occurs in the first versus the second year of the window; the two single-boom paths are distinguished as transformative (a) and (b) below.

Investment paths during the window. Recall that I_t is chosen at date t after observing z_t and determines K_{t+1} via the law of motion (2). Throughout the window, the optimal capital stock moves according to Proposition 3.1: each realized boom contributes the factor e^{ξ} in (11), and each shift in the boom probability contributes the factor $(\Omega(p_t)/\Omega(p_{t-1}))^{1/(1-\alpha)}$. When p rises to $p_{\max}$ in 2028, the first affected investment decision is I_{2028}: chosen in 2028 using the option value of $p_{\max}$ but before the 2029 boom outcome is known. This investment determines K_{2029}.

2028 investment: same for all scenarios. Write $K^*(z, p)$ for the optimal capital stock (9) at productivity z and boom probability p; z_0 denotes pre-boom AI productivity. With $p_{\max} = 0.50$, the optimal capital stock entering 2029 rises from $K^*(z_1, 0) = \$2{,}212\text{B}$ to $K^*(z_1, p_{\max}) = \$4{,}680\text{B}$, where $z_1 = z_0 e^{\xi}$ is the post-initial-boom productivity. The rise is the option-value term of Proposition 3.1: p shifts from approximately zero to $p_{\max}$ with no boom yet realized. By the calibration, $K^*(z_1, 0)$ is the capital stock entering 2028. The required investment is $I_{2028} = K^*(z_1, p_{\max}) - (1 - \delta_a)K^*(z_1, 0) = \$3{,}021\text{B}$. This is identical across the four scenarios because the 2029 boom has not yet been observed.

2029 investment: outcomes diverge. The 2029 boom outcome is now known. If no boom occurred (moderate and transformative (b)), z remains at z_1 and investment is just depreciation: $I_{2029} = \delta_a K^*(z_1, p_{\max}) = \$1{,}170\text{B}$. If a boom occurred (transformative (a) and singularity), z jumps to $z_2 = z_0 e^{2\xi}$ and the capital stock must adjust: $I_{2029} = K^*(z_2, p_{\max}) - (1 - \delta_a)K^*(z_1, p_{\max}) = \$12{,}770\text{B}$.

2030 investment: outcomes diverge again. Both the 2030 boom outcome and the reversion of p to p_* are reflected. Because $\Omega(p_*)$ is substantially below $\Omega(p_{\max})$, the optimal capital stock falls in scenarios where no new boom occurred. If a late boom raised z, investment remains elevated.

2031: investment replaces depreciation unless another boom occurs. In the absence of a further boom, annual investment equals $\delta_a K^*$: \$586B (moderate), \$2,037B (transforma-

tive), and \$7,087B (singularity). A new boom raises the optimal capital stock and requires additional investment.

Partial adjustment. While this describes a frictionless investment sequence, in reality there are substantial frictions. Ideally, one would micro-found these with a model of adjustment costs, for example, assuming the standard quadratic form (Hayashi, 1982, Jorgenson, 1963). However, such a model would predict that investment after a single shock follows a decelerating path regardless of the adjustment speed, whereas the data show acceleration. An interesting question for further research is how to model adjustment costs in the AI industry, where manufacturing lead times and power infrastructure are first-order concerns. Therefore instead we use the considerations above to determine an optimal K^* and assume that the capital stock follows $K_{t+1} = K_t + (1/\theta_a)(K^* - K_t)$, with $\theta_a = 3$ (one-third of the gap closed per year), beginning with the 2028 investment decision, so that the first partially adjusted stock is K_{2029}. Prior to 2028, we take investment from the data. The path starts in 2028 from the calibrated target $K^*(z_1, 0)$, which equals the capital stock implied by the data, so the gap that opens in 2028 is due to the rise in the boom probability alone. Figure 1 displays the resulting paths, whereas Figure C.1 shows the corresponding paths with no adjustment frictions.

Calibrating p_* to aggregate valuations. What happens following 2030 is necessarily more speculative. We calibrate a long-run boom probability p_* to the aggregate equity-market price-to-dividend ratio, taking the boom size as given. As of May 2026, the S&P 500 trades at a price-to-dividend ratio of approximately 94, equivalent to a dividend yield of 1.06%. Combining the 30-year TIPS yield of 2.7% with a 5% real equity premium gives a real discount rate $r = 7.7\%$, so the Gordon decomposition $D/P = r - g$ implies a long-run real dividend growth rate of 6.6%. Because dividends are by construction the cash flows

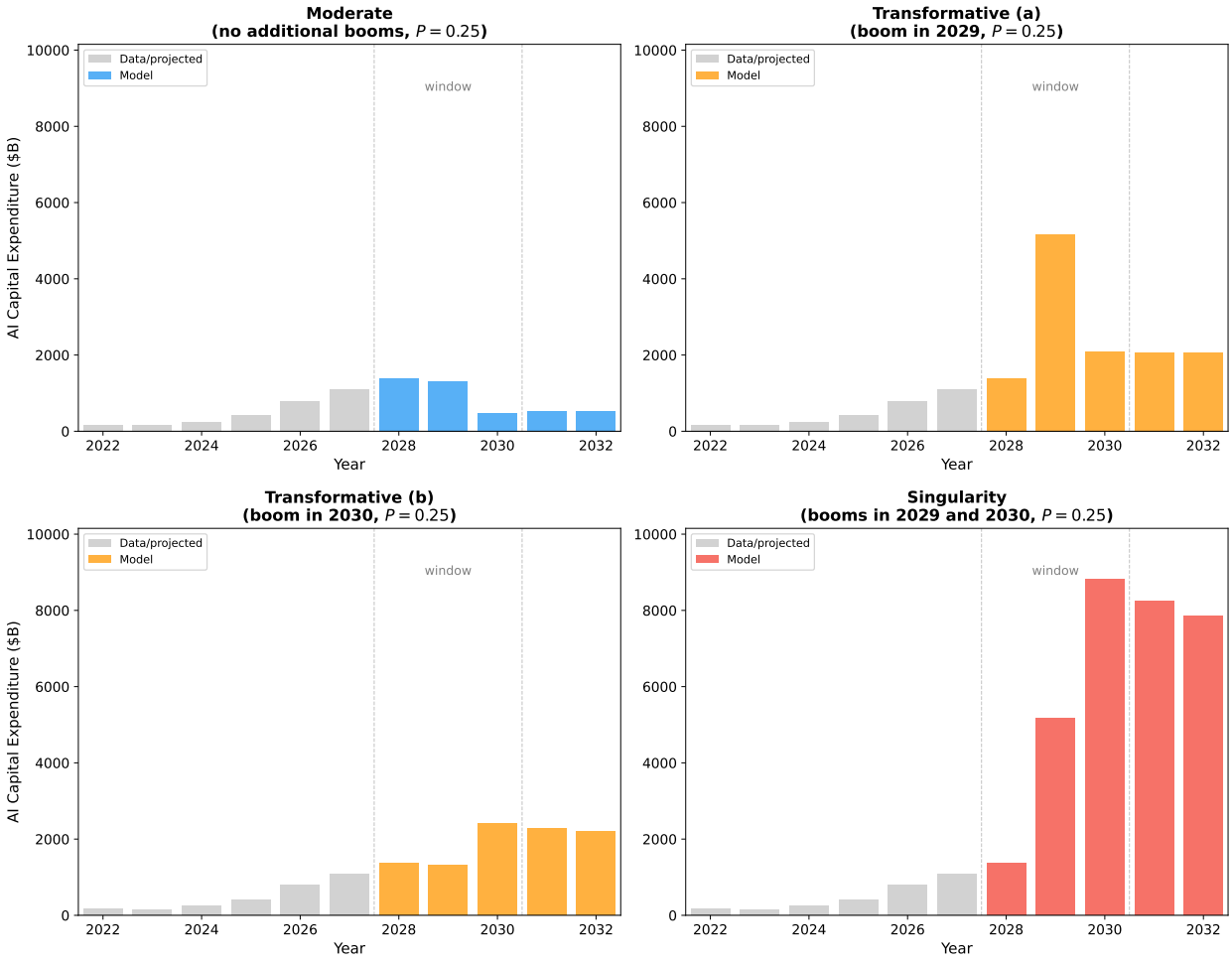


Figure 1: Investment paths by scenario. Gray bars show actual and projected AI capital expenditure through 2027 (Table 1). Colored bars show investment implied by partial adjustment toward the frictionless target K^*: each year the capital stock closes a fraction $1/\theta_a$ of the gap, with $\theta_a = 3$, beginning with the 2028 response to the elevated boom probability. Dashed lines delineate the high-probability window (2028–2030). Investment in 2028 is identical across all four panels because it is chosen before the first window boom is observed. The calculation assumes no additional booms past 2030.

that accrue to shareholders—net of all investment, including the expansionary investment that scales capital up after each productivity boom—this calculation asks only at what rate aggregate cash flows compound.[11] Equating this growth rate to the model-implied AI-sector growth $p(e^{\xi} - 1)$ gives $p = 0.027$. We set

$$p_* = 0.03 \tag{15}$$

for the remainder of the paper. At this value, expected AI-sector growth is 7.4% per year.

Two caveats apply to this calculation. The growth rate on the right-hand side of the Gordon decomposition is the growth rate of aggregate dividends, while $p_*(e^{\xi} - 1)$ is the growth rate of the AI sector. First, in the moderate scenario, the AI share remains small for decades, and aggregate growth remains well below $p_*(e^{\xi} - 1)$. Justifying the observed price–dividend ratio would then require either a larger p_* or spillovers from the AI sector to the rest of the economy, which the model excludes. Second, part of aggregate dividend growth would occur without AI booms. Subtracting it lowers the implied p_*.

Table 5 reports the post-window installed capital stocks K_a^* implied by the optimality condition (9) at $p = p_*$, together with the corresponding capacity in gigawatts at a working conversion of $40 billion per gigawatt.[12]

[11] The investment data used to calibrate ξ are nominal, but the inflation adjustment over 2024–2027 is small relative to the orders of magnitude involved.

[12] Van Nieuwerburgh (2026) calculates a cost of $40 billion per GW; Amodei (2026) estimates $10–$15 billion. In what follows, we compare forecasts of power demand with those implied by our model. At a lower cost per GW, the capacity implied by a given dollar investment is correspondingly larger. Implications for GDP, however, depend on whether productivity is conceived of in units of GW versus units of dollar investment. Van Nieuwerburgh gives a forecast of 200 GW, which at $40 billion per GW is $8 trillion of installed capital, close to the transformative scenario.

Table 5: Post-window installed AI capital and implied capacity

Scenario	Installed capital K_a^* (\$B)	Implied capacity (GW)
Moderate	2,343	59
Transformative	8,149	204
Singularity	28,347	709

Notes: Installed capital is the post-window optimal stock implied by (9) at $p = p_*$, with productivity at its scenario-terminal value and no further booms realized after 2030. Implied capacity converts dollars to gigawatts at a working value of \$40 billion per GW (Van Nieuwerburgh, 2026); Amodei (2026) estimates \$10–\$15 billion, and the GW figures scale inversely with this cost.

3.4 Discussion

The revealed-preference argument identifies the productivity boom that managers believe has occurred. It does not establish that the boom has in fact occurred. Two channels could drive a wedge between the implied $\hat{\xi}$ and the true productivity gain.[13]

First, the investment may simply reflect a bubble. Managers are not immune to collective over-optimism, and the history of technology investment is replete with episodes—the late-1990s fiber-optic buildout, for instance—in which firms invested heavily in capacity that was not justified by subsequent demand. How exactly this scenario would play out in the case of AI is not known, but it is possible, for example, that the productivity-enhancing power of AI may be a mirage. In this case, customers would be unwilling to pay for more AI usage. In terms of the model, this is essentially a mis-estimation of ξ in (3), resulting in negative NPV investments, declining stock valuations, failure to make interest payments on debt, and perhaps bankruptcy. The resulting disinvestment would then lower the GDP growth estimates in Section 4.

[13]Several technical considerations affect the size of the calibrated boom. As noted in the discussion following Table 1, calibrating to the full private-sector AI investment universe would raise the boom size and the installed capital. Similarly, the 2022 baseline encompasses non-AI uses; to the extent that some fraction represents non-AI corporate capital, the true pre-boom AI capital stock is smaller and the calibrated boom size correspondingly larger. However, if some of the investment is anticipatory (namely in advance of upward shifts in p_t), then this would lower the boom size. A lower value of depreciation δ_a also lowers the boom size (see Appendix B for details).

Second, the structure of the AI infrastructure market may lead to over-investment. Dasgupta and Stiglitz (1980) show that in R&D races with winner-take-all payoffs, each firm invests as if it may capture a disproportionate share of the market, producing aggregate investment in excess of the cooperative optimum. The current AI buildout has this character: the five firms in Table 1 are simultaneously building infrastructure to serve overlapping markets, and each firm's capital expenditure may reflect an expectation of capturing a larger share than is collectively feasible. If so, aggregate investment overstates the productivity gain that any single consistent set of beliefs would imply.

To this point, suppose, however, that each firm were to act as if it will be the sole survivor in 2027, with no strategic interaction at all and with all investment bespoke to that firm's own AI program. The first-order condition then applies separately to each firm's own capital stock, and the revealed-preference calibration delivers a firm-specific $\hat{\xi}_i$, with $e^{\hat{\xi}_i} = K_{i,2028}/K_{i0}$. Table 6 reports this calibration, carried out on the per-firm capex values in Table 1; $\hat{\xi}_i$ ranges from 0.89 (Amazon) to 2.00 (Oracle). Thus the implied productivity jump is high even when considered on a per-firm basis. That said, if in fact not all firms survive, then installed capital will be lower, reducing GDP growth in what follows.

Table 6: Per-firm calibration of $\hat{\xi}_i$

Firm	$I_{i,\text{pre}}$ (\$B)	$K_{i,2028}$ (\$B)	$e^{\hat{\xi}_i}$	$\hat{\xi}_i$
Amazon	64	622	2.43	0.89
Meta	32	395	3.08	1.13
Alphabet	31	507	4.09	1.41
Microsoft	25	482	4.82	1.57
Oracle	7	207	7.40	2.00
Aggregate	159	2,212	3.48	1.25

Notes: Pre-boom investment $I_{i,\text{pre}}$ is firm i's 2022 capital expenditure from Table 1. The capital stock entering 2028, $K_{i,2028}$, applies (12) to firm i's annual investments from Table 1, starting from the firm's own pre-boom steady state $K_{i0} = I_{i,\text{pre}}/\delta_a$ with $\delta_a = 0.25$. The boom size is $e^{\hat{\xi}_i} = K_{i,2028}/K_{i0}$, as in Section 3.3.

More broadly, just as over-optimism about both the technology and one's own prospects could lead to over-investment, managerial risk aversion, anchoring to the status quo, or the desire for the quiet life could lead to under-investment. Our revealed-preference argument would then be conservative.[14] Some industry forecasts align most closely with the singularity scenario, or perhaps slightly above it, implying that future booms are viewed not as the high end but as the median, and that future investment will dwarf current plans. For example, Musk (2026b) forecasts "at least, five years from now, a few hundred gigawatts per year of AI in space and rising," speaking in early 2026. The singularity scenario implies roughly $28 trillion of installed capital, or roughly 700 GW of installed capacity (Table 5). Musk's forecast, assuming $\delta_a = 0.25$, calls for a steady-state stock of four times "a few hundred GW"—on the order of 1,000 GW, the same order of magnitude as the singularity scenario. Consistent with this public statement, SpaceX's S-1 (Space Exploration Technologies Corp., 2026), filed in May 2026, sets a goal of launching 100 GW of orbital AI compute capacity per year, with first deployments "as early as 2028." Sustained, this launch flow corresponds at $\delta_a = 0.25$ to a steady-state installed stock of 400 GW—by itself twice the sector-wide stock of the transformative scenario in Table 5. The filing's goal, moreover, concerns a single firm, and one outside the five-firm universe of Table 1 (Appendix A.3 treats SpaceX's AI segment as a sixth hyperscaler). Assuming SpaceX would not be the sector's only provider of data-center capacity, the implied sector-wide stock exceeds 400 GW, more than half the singularity scenario.

As a second example, Amodei (2024, 2026) envisions "millions of instances" of AI "smarter than a Nobel Prize winner" running at "10x–100x human speed"—a "country of geniuses in a datacenter"—arriving in roughly 2028. The model accommodates this scenario directly:

[14] A related mechanism operates through the market for distressed assets: if firms anticipate acquiring rivals' capital at below replacement cost should those rivals fail, the option to buy later reduces the incentive to build now. In this case, the productivity gains estimate that we find would be lower than what would make investment optimal.

hold A_a fixed, and let the country of geniuses enter as an addition to effective labor, so that $z_a = A_a^{1/(1-\alpha)}\bar{L}_a$ (footnote 5) rises in proportion to $\bar{L}_a$. Under one interpretation of Amodei's numbers—10 million instances at 30 times human speed—the geniuses add 300 million effective workers, arriving in 2028 on top of the AI sector as it stands at the end of 2027. That baseline already reflects the initial boom. Because effective labor paid a common wage is allocated in proportion to sector output, the sector's effective workforce is 3.2% of the 167 million U.S. labor force before the initial boom (Table 7), or 5.3 million; the initial boom multiplies this by $e^{\hat{\xi}} \approx 3.5$, to 18 million by end-2027.[15] The arrival of 300 million geniuses in 2028 would then multiply $\bar{L}_a$, and with it z_a, by a further factor of roughly 17—approximately two booms of size $\hat{\xi}$ beyond the one that has already occurred ($\log 17 \approx 2.8$, or $2.3\hat{\xi}$). Since the model's window allows for at most two additional booms—the singularity scenario—Amodei's description sits, under this interpretation, at roughly the singularity scenario.

Ultimately, the model's predictions should be read as conditional on the revealed-preference estimate of $\hat{\xi}$, which is itself conditional on the assumption that the firms driving the buildout are, in aggregate, not destroying value.

4 GDP growth

The AI sector's share of total output, denoted s_a, depends on ξ, r_a, and the long-run boom probability p_*. Using (1):

$$s_a \equiv \frac{Y_a}{Y_a + Y_n} = \frac{z_a^{1-\alpha}(K_a^*)^\alpha}{z_a^{1-\alpha}(K_a^*)^\alpha + z_n^{1-\alpha}(K_n^*)^\alpha}. \tag{16}$$

[15] Under the calibration of Section 3.3, capital has adjusted to the initial boom by end-2027, so that $Y_a \propto z_a$.

Pre-boom, the AI sector accounts for approximately 3.2% of output. Table 7 reports, for each scenario, the post-window AI output share, the cumulative increase in GDP, and the steady-state level of AI investment.[16]

Table 7: Permanent GDP gains and AI sector outcomes: post-window steady state

	AI share (%)	Permanent GDP gain (%)	Post-window I_a ($B/yr)
Pre-boom	3.2	—	159
Moderate	10.4	8.1	586
Transformative	28.7	35.9	2,037
Singularity	58.4	132.6	7,087

Notes: Each row evaluates the post-window steady state for one scenario of Table 4, with productivity at its scenario-terminal value, the boom probability at $p_* = 0.03$, and no further booms realized after 2030. "AI share" is the AI sector's share of output s_a, defined in (16). "Permanent GDP gain" is the increase in the level of GDP that remains after capital has fully adjusted, relative to the pre-boom economy, holding non-AI output fixed and excluding normal-times growth. "Post-window I_a" is steady-state AI investment, which replaces depreciation ($\delta_a K_a^*$). As booms continue to arrive at rate p_*, the AI share and GDP contribution grow further (Figure 3).

The moderate scenario's permanent GDP gain has a simple interpretation. AI initially accounts for approximately 3.2% of GDP, and its output becomes roughly 3.5 times as large once capital has adjusted. With non-AI output held fixed, aggregate GDP therefore rises by approximately $3.2\% \times (3.5 - 1) = 8\%$.[17] This is a permanent increase in the level of GDP: with no further booms, annual output remains approximately 8% above its pre-boom level after capital adjustment, excluding normal-times growth.

We now translate the calibrated scenarios into GDP paths. At each date t, AI output is $Y_{at} = z_{at}^{1-\alpha} K_{at}^{\alpha}$, where z_{at} reflects the cumulative booms realized in that scenario. Non-

[16] The post-window steady state is the economy with productivity at the scenario's terminal value, capital fully adjusted to $K^*(z_a, p_*)$, and no further booms after 2030, so that investment replaces depreciation and nothing changes from year to year. Because booms continue to arrive at rate p_*, it is the steady state conditional on no further booms, not the expected long-run outcome; Figure 3 shows the share and GDP continuing to grow.

[17] In the calibration, AI output rises from $763.2B to $2,706.0B after capital adjusts to $K^*(z_a, p_*)$. Relative to pre-boom GDP of $24,116.9B, the gain is $(2{,}706.0 - 763.2)/24{,}116.9 = 8.06\%$.

AI output is held at its pre-boom level Y_{n0}.[18] Cumulative GDP growth from AI is $(Y_{at} + Y_{n0})/\text{GDP}_0 - 1$, where $\text{GDP}_0 = Y_{a0} + Y_{n0}$ is pre-boom GDP.

GDP during the transition. Figure 2 shows cumulative GDP growth from AI for each scenario during the transition and its immediate aftermath (2024–2032). In the moderate scenario, the permanent 8% gain is already realized by 2030. GDP is temporarily about 10% above its pre-boom level and then falls back toward the permanent gain: the capital built during the window, when the boom probability was elevated, exceeds the long-run optimum once no further boom occurs, and the excess depreciates. In the transformative scenario, the second boom raises GDP by about 35% by 2030 and 36% once capital has fully adjusted. In the singularity scenario, two additional booms compound to raise GDP by about 84% by 2030 and 133% once capital has fully adjusted.[19] Post-window AI investment is approximately \$586B/yr (moderate), \$2,037B/yr (transformative), and \$7,087B/yr (singularity); the last of these would exceed current U.S. gross private fixed investment.

Long-run paths (2030–2050). Figure 3 extends the paths to 2050, allowing for ongoing booms. Starting from each scenario's post-window productivity level z, booms arrive i.i.d. at rate $p_* = 0.03$ per year. Each boom multiplies z by e^{ξ}. Under (14), $p_{2030} = p_*$ governs the boom of 2031, so there are 20 draws by 2050, and the number of booms realized by year t is binomial with parameters $t - 2030$ and p_*. Capital continues to follow the partial-adjustment rule of Section 3.3, closing a fraction $1/\theta_a$ of the gap to $K^*(z_{at}, p_*)$ each year, so that output lags each boom as it does during the window. We compute the expected paths and percentile

[18] Holding Y_n fixed isolates the AI contribution to growth. The shares reported below are therefore the AI sector's share of an AI-only-growing economy, not its share of calendar-year GDP after incorporating ordinary non-AI growth; over multi-decade horizons the latter would be smaller, most noticeably in the moderate scenario where the AI contribution accrues slowly.

[19] In a May 2026 interview (Musk, 2026a), Musk forecasts that "digital intelligence will exceed the sum of all human intelligence" within five years, and that "the economy is probably twice its current size in 5, maybe 6 years"—a doubling of GDP by 2031, against the singularity scenario's approximately 133% fully adjusted GDP increase.

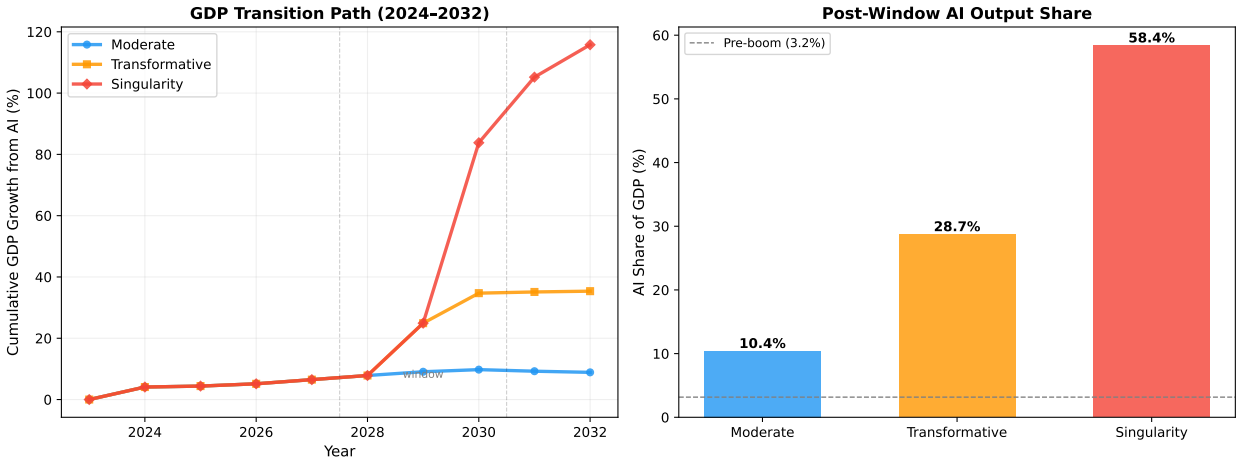


Figure 2: GDP transition paths. Left panel: cumulative GDP growth from AI during the transition and its immediate aftermath (2024–2032), using data-anchored partial adjustment ($\theta_a = 3$). The transformative line is path (a), with the additional boom in 2029. Dashed lines mark the 2028–2030 window. Right panel: post-window AI output share of GDP; the dashed line shows the pre-boom share (3.2%).

bands exactly, by enumerating the 2^{20} possible boom histories with their probabilities.[20]

The compounding effect of post-window booms is first order. In the moderate scenario, expected cumulative GDP growth reaches approximately 40% by 2050, with the AI sector's expected output share rising to roughly 22% (10th–90th percentile ranges: 8–114% for GDP, 10–55% for share). In the singularity scenario, expected cumulative GDP growth reaches 522% by 2050, with the AI sector's expected share reaching 71% of output (10th–90th percentile ranges: 133–1,415% for GDP, 58–94% for share). The bottom row of the figure provides a detailed view of the moderate scenario.[21]

[20] Appendix C reports the same figure with capital at its optimal level each year from 2030 on. Expected growth by 2050 is then 44%, 161%, and 567% in the three scenarios, rather than 40%, 148%, and 522%.

[21] As a comparison, SpaceX's May 2026 S-1 (Space Exploration Technologies Corp., 2026) states a quantified AI-related total addressable market of $26.5 trillion, comprising $2.4T in AI infrastructure (RAND's 2030 forecast of 235 GW global data-center demand × 70% AI share, marked to a $3.33/GPU-hour neocloud rental rate), $760B in consumer subscriptions, $600B in digital advertising, and $22.7T in "enterprise applications," which is the Digital Cooperation Organization's estimate of the entire 2026 digital economy. To put these values into perspective, expected Y_{at} first crosses $10T around 2030 in the singularity scenario, 2032 in the transformative scenario, and 2050 in the moderate scenario; $30T is crossed around 2032, 2048, and 2065, respectively.

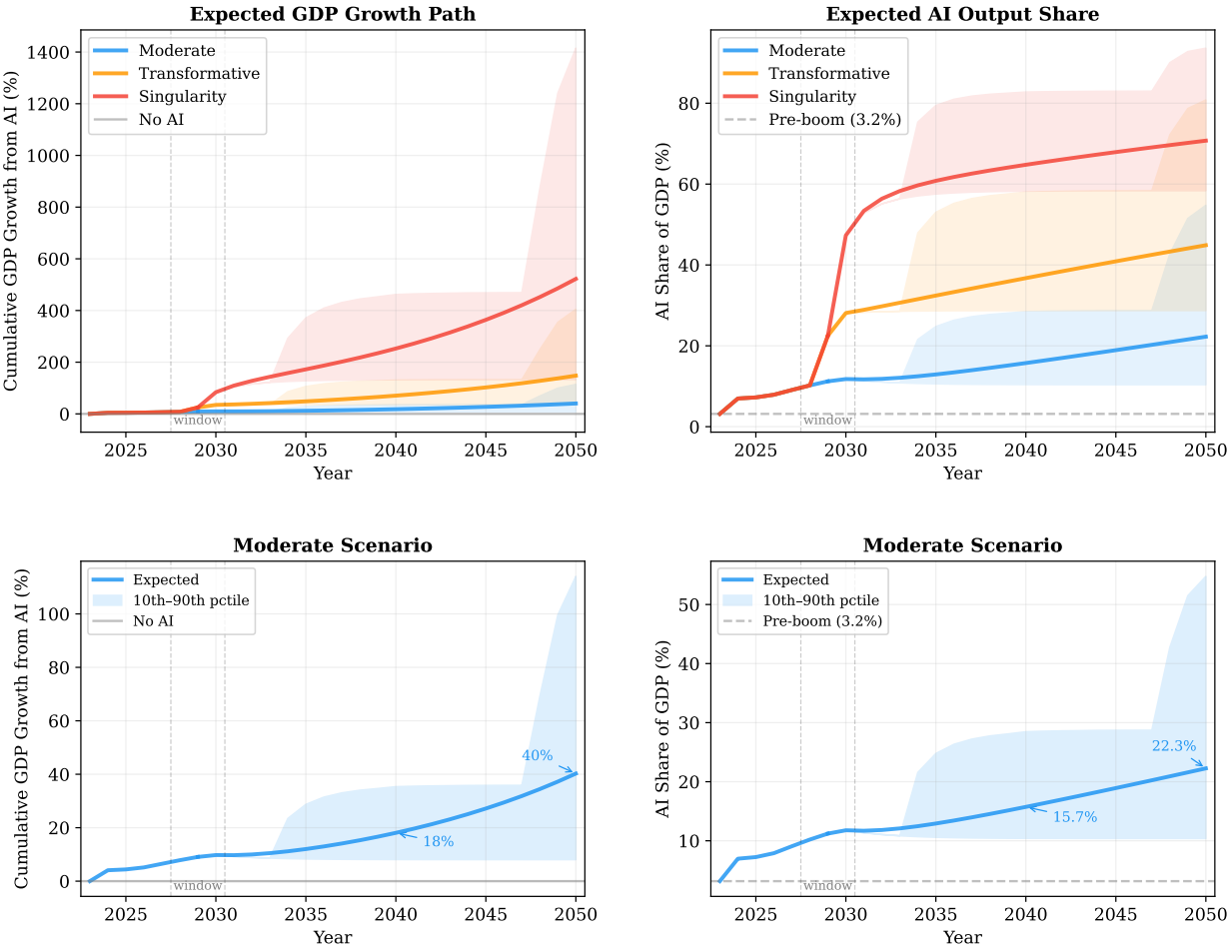


Figure 3: GDP growth distributions (2024–2050). Top row: all three scenarios. Bottom row: the moderate scenario alone. Solid lines show the expected path; shaded bands show the 10th–90th percentile range, computed exactly over the 2^{20} post-window boom histories. Gray lines show the no-AI baseline. The paths use observed and projected investment through 2027. Beginning with the 2028 investment decision, capital follows the partial-adjustment rule of Section 3.3 ($\theta_a = 3$). The transformative scenario uses path (a); from 2031 onward booms arrive i.i.d. at rate $p_* = 0.03$. Vertical dashed lines mark the 2028–2030 window.

Growth rates in the very long run. Three observations summarize this discussion. First, even the moderate scenario—no additional boom during the elevated-probability window—delivers approximately 8 percentage points of additional GDP once capital has adjusted and 40 percentage points in expectation by 2050, an order of magnitude above task-based estimates. Second, because the model assumes no spillovers from the AI sector to the non-AI sector, the size of the AI sector after the transition is of first-order importance: aggregate growth from AI is the sector's own growth weighted by its share, and as the share approaches one, aggregate growth converges to the AI sector's growth rate. The gap between a post-window share of 10% (moderate) and 58% (singularity) is what compounds into the difference between 40% and 522% expected growth by 2050. Third, once the AI share is large, the Bernoulli uncertainty over boom arrivals drowns out every other source of variation: in the singularity scenario, the 10th–90th percentile range spans 133–1,415% of cumulative growth by 2050 (Figure 3). The width of this fan, rather than any single path, is the model's message about the long run.

The terawatt of AI compute capacity that recurs in industry discussion (Section 3.4) can be put in perspective using these results. At the working conversion of Table 5, one terawatt of installed capacity exceeds the post-window capital stock of the singularity scenario (709 GW). The capacity that industry leaders describe building thus exceeds the capital stock of the scenario in which the third observation applies with full force: expected AI-sector output growth of 7.4% per year, together with uncertainty that comes to dominate the expected path.

5 Implications for the equity premium and for interest rates

The open-economy model of Sections 3–4 treats the required return r_a as exogenous. This is the natural formulation for an economy in which global investors supply capital elastically, and it is the most transparent framework for mapping investment decisions into the size of the productivity boom. Moreover, the mapping from risks into interest rates and equity premia is not at present well-understood, as most well-disciplined models fail in various regards (see, e.g., Guo and Wachter, 2025). Nonetheless, the model has strong implications for interest rates and risk premia within the standard framework. We discuss these here.

5.1 Preferences

A representative agent has recursive preferences of the form introduced by Epstein and Zin (1989) with unit elasticity of intertemporal substitution, risk aversion γ, and discount factor β. When $\gamma = 1$, this reduces to time-separable log utility; when $\gamma \neq 1$, the EIS remains unity but the agent is more (or less) risk averse than the log case.

Let W_t denote the cum-dividend value of the consumption claim, and $P_t = W_t - C_t$ its ex-dividend price. With unit EIS, the price–consumption ratio is constant: $P_t/C_t = \beta/(1-\beta)$, so the return on wealth is $R_{W,t+1} = \beta^{-1}\, C_{t+1}/C_t$. The stochastic discount factor is (see Miller et al., 2025)

$$M_{t+1} = \beta\, \mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{1-\gamma}\right]^{-1}\left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}. \tag{17}$$

The expectation term normalizes the SDF so that $\mathbb{E}_t[M_{t+1}R_{W,t+1}] = 1$. It introduces γ even though the wealth–consumption ratio is constant, and is the source of precautionary savings demand.

Aggregate consumption growth is

$$\Delta c_{t+1} = \mu + \sigma\,\varepsilon_{t+1} + x_{d,t+1} + x_{b,t+1}, \tag{18}$$

where $c_t = \log C_t$, μ and σ are the mean and volatility of normal-times growth, $x_{d,t+1}$ is a rare disaster that equals $\xi_d < 0$ with probability p_d and zero otherwise, and $x_{b,t+1}$ is a rare boom that equals $\xi_b > 0$ with probability p_b and zero otherwise. The three shocks are mutually independent and i.i.d. over time, with $\varepsilon_{t+1} \sim N(0,1)$. Following Barro (2006) and Wachter (2013), we set $p_d = 0.036$ and $\xi_d = -0.15$. The following subsection discusses the corresponding objects for booms.

5.2 Assumptions on aggregate cash flows

Equation 18 requires the jump ξ_b in aggregate consumption, not output. Mapping the AI-sector productivity boom ξ into a consumption boom involves two considerations. First, the boom is sector-specific: when AI productivity jumps by e^{ξ}, non-AI output is unchanged, so the aggregate effect depends on the AI sector's output share s_a (16).

Second, even if the boom applied to the entire economy, short-run investment dynamics would create a wedge between aggregate output and aggregate consumption. It is for this reason that Gourio (2012) assumes that disasters affect productivity and the capital stock proportionally, and that Miller et al. (2025) assume that $\alpha = 1$ and that the shock applies to capital. These assumptions lead to a production economy that is isomorphic to the endowment economy, greatly simplifying the computation of consumption growth, the stochastic discount factor, and thus investment. Under these assumptions the immediate effect of the boom on consumption would be the same as on productivity and capital.[22]

[22] Wealth in the economy equals undepreciated capital plus current output, $(1-\delta)K_t + Y_t$. Specializing to $\alpha = 1$, production is linear in capital, $Y_t = A_t K_t$, where we write productivity as A_t rather than $z_t^{1-\alpha}$, following the notation of Miller et al. (2025), so that $W_t = (1 - \delta + A_t)K_t$. If C_t equals total consumption,

As discussed in Section 3, we do not make either assumption in part because they are not realistic in this setting, and also (relatedly) because we are using the unanticipated shock to productivity to calibrate the investment response. However, a full solution, which would incorporate the gradual growth of the AI sector, as well as complicated short-run consumption dynamics, is difficult, and would not appreciably add to the intuition we are attempting to convey.

We therefore consider two versions of an endowment economy as a means of showing the range of effects the model can produce. We first hold the AI sector's output share fixed at its post-window value. The effect of post-window productivity booms is then confined to this portion of the economy. We set

$$\xi_b \equiv s_a \cdot (1-\alpha)\,\xi, \tag{19}$$

where s_a is the post-window AI output share of each scenario (Table 7). This equation is an approximation even when the AI sector is in fact frozen in time.

The second version takes (19) and sets $s_a = 1$. This also understates the long-run effect on consumption (which will grow at the same rate as output, namely at ξ). However, even this level produces dramatic effects on the risk-free rate and equity premium.

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then the law of motion implies that $K_{t+1} = W_t - C_t$ and hence the capital accumulation equation

$$K_{t+2} + C_{t+1} = (1-\delta)K_{t+1} + A_{t+1}K_{t+1}$$

can be equivalently written as

$$W_{t+1} = (W_t - C_t)\big(1 - \delta + A_{t+1}\big),$$

and thus $1 - \delta + A_{t+1}$ is the return on invested wealth. The assumption of unit EIS then implies that

$$\frac{C_{t+1}}{C_t} = \frac{W_{t+1}}{W_t} = \frac{1-\delta+A_{t+1}}{1-\delta+A_t}\frac{K_{t+1}}{K_t}.$$

In Miller et al. (2025), for example, A_t is a constant and K_{t+1} is hit by the rare event. Gourio (2012) shows a more general result that states that if capital and productivity fall by the same amount, then all quantities scaled by productivity are independent of the realization of the rare event.

5.3 Asset prices

The risk-free rate and equity premium follow from the Euler equation $\mathbb{E}_t[M_{t+1}R_{t+1}] = 1$. For a risk-free bond with gross return R_f, $R_f^{-1} = \mathbb{E}_t[M_{t+1}]$. The Euler equation implies that the price–consumption ratio equals $\beta/(1-\beta)$, and hence, if we let $R_{W,t+1} = (P_{t+1} + C_{t+1})/P_t$ be the gross return on wealth, $R_{W,t+1} = \beta^{-1}\, C_{t+1}/C_t$.

Substituting (17) into the Euler equation and defining

$$\Phi_d(u) = (1 - p_d) + p_d\, e^{u\,\xi_d}, \qquad \Phi_b(u) = (1 - p_b) + p_b\, e^{u\,\xi_b} \tag{20}$$

implies

$$r_f = \rho + \mu + \frac{1 - 2\gamma}{2}\,\sigma^2 + \log\frac{\Phi_d(1-\gamma)}{\Phi_d(-\gamma)} + \log\frac{\Phi_b(1-\gamma)}{\Phi_b(-\gamma)}, \tag{21}$$

where $\rho = -\log\beta$ and $r_f = \log R_f$.

Taking logs of $\mathbb{E}_t[R_{W,t+1}]$ and subtracting $r_f = \log R_f$ yields the (log) equity premium:

$$\log\mathbb{E}[R_W] - r_f = \gamma\,\sigma^2 + \log\frac{\Phi_d(1)\,\Phi_d(-\gamma)}{\Phi_d(1-\gamma)} + \log\frac{\Phi_b(1)\,\Phi_b(-\gamma)}{\Phi_b(1-\gamma)}. \tag{22}$$

For small p_d and p_b, this premium can be approximated as:

$$\log\mathbb{E}[R_W] - r_f \approx \gamma\,\sigma^2 + p_d(e^{\xi_d} - 1)(1 - e^{-\gamma\xi_d}) + p_b(e^{\xi_b} - 1)(1 - e^{-\gamma\xi_b}).$$

While we use (22) for all calculations below, the approximation highlights the source of risk: both disasters and booms raise the equity premium.

Table 8 reports the risk-free rate and equity premium for the pre-AI economy, for the three post-window scenarios, and for the absorbing limit. Figure 4 plots the same quantities as functions of risk aversion γ.

Several features stand out. First, the range across columns is wide, reflecting the difficulty

Table 8: Equilibrium rates of return

	No AI	Moderate	Transformative	Singularity	Absorbing limit
Boom probability p_b	0	0.03	0.03	0.03	0.03
Aggregate boom size ξ_b	—	0.086	0.239	0.485	0.831
Panel A: Risk-free rate r_f (%)					
$\gamma = 1$	3.90	4.15	4.54	5.06	5.61
$\gamma = 3$	3.63	3.84	4.03	4.07	3.96
$\gamma = 5$	3.29	3.47	3.54	3.46	3.36
Panel B: Equity risk premium (%)					
$\gamma = 1$	0.12	0.14	0.28	0.81	2.22
$\gamma = 3$	0.39	0.45	0.80	1.80	3.87
$\gamma = 5$	0.72	0.82	1.28	2.41	4.47

Notes: Entries are the equilibrium risk-free rate r_f and the equity risk premium on the claim to aggregate consumption (22), in annual percent, at risk aversion $\gamma \in \{1, 3, 5\}$; the elasticity of intertemporal substitution equals one throughout. The aggregate boom size ξ_b is not a free parameter: $\xi_b = s_a(1-\alpha)\xi$ scales the calibrated AI-sector boom $\xi \approx 1.25$ by each scenario's post-window AI output share s_a (Table 7); the absorbing limit sets $s_a = 1$, which obtains after sufficiently many booms. Booms arrive with probability $p_b = p_* = 0.03$ per year. Remaining parameters: rate of time preference $\rho = 0.02$; mean consumption growth $\mu = 0.025$ with volatility $\sigma = 0.02$; disasters of size $\xi_d = -0.15$ arrive with probability $p_d = 0.036$.

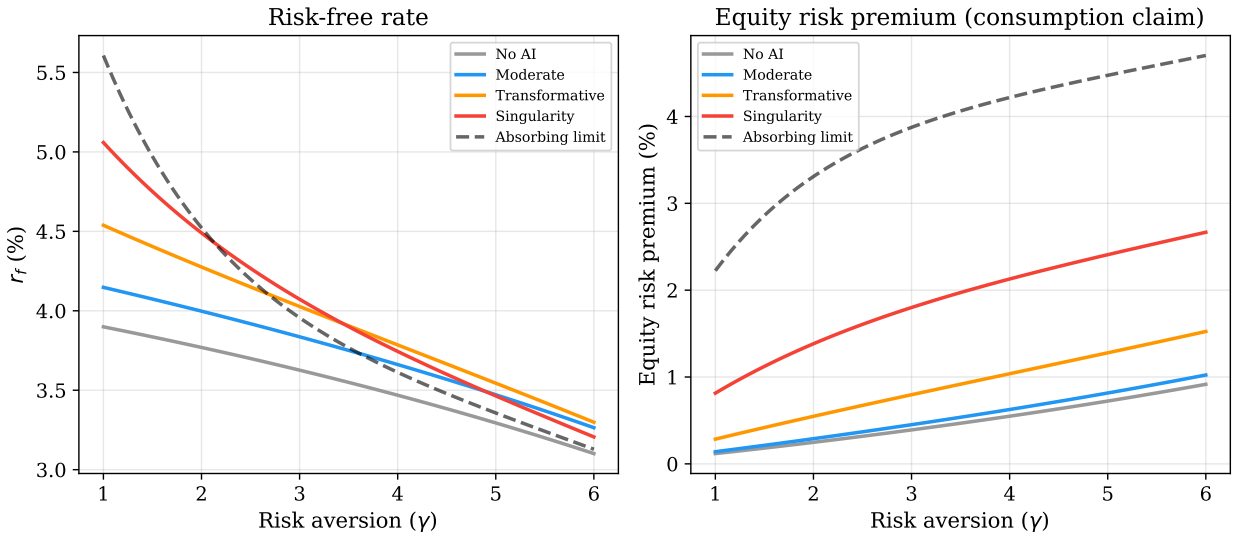


Figure 4: Equilibrium rates of return. Left panel: risk-free rate. Right panel: equity risk premium on the claim to aggregate consumption. Gray line shows the pre-AI economy ($p_b = 0$); colored lines show the three post-window scenarios with $p_b = p_* = 0.03$ and scenario-specific ξ_b (Table 8); dashed black line shows the absorbing limit $\xi_b \approx 0.831$. Other parameters: $\rho = 0.02$, $\mu = 0.025$, $\sigma = 0.02$, $p_d = 0.036$, $\xi_d = -0.15$.

of mapping a sectoral shock to aggregate asset prices without solving the full non-stationary model. At $\gamma = 3$, the equity premium increase ranges from 6 bp (moderate) to 348 bp (absorbing limit). The scenario columns treat the AI sector as permanently small; the absorbing limit treats it as if it already dominates the economy. Over time, the economy transitions from the former toward the latter as the AI sector's output share grows. Both calculations scale the consumption shock by $1-\alpha$, reflecting the immediate output response before capital adjusts. Using the fully adjusted response would imply a larger consumption shock and a higher equity premium in the same endowment model.

Second, at $\gamma = 3$, the equity premium rises from 0.39% without AI to 0.45% in the moderate scenario, 0.80% in the transformative scenario, and 1.80% in the singularity scenario. Third, the risk-free rate rises in all cases, reflecting higher expected consumption growth: agents who expect to be richer tomorrow save less today. Fourth, the equity premium also rises, because the boom creates a state in which both consumption and equity returns are

high. With unit EIS, the SDF is $M_{t+1} \propto \exp(-\gamma\,\Delta c_{t+1})$, so any $\gamma > 0$ penalizes states with high consumption growth. The positive covariance between boom-driven consumption growth and equity returns raises the required return on equity for all levels of risk aversion reported in the table. Fifth, the effect on the risk-free rate is decreasing in γ while the effect on the equity premium is increasing—higher risk aversion strengthens precautionary savings (damping the rise in r_f) and raises the price of risk (amplifying the rise in the equity premium).

5.4 Sovereign default and the observed yield

The general equilibrium risk-free rate in Table 8 is a rate on a default-free claim. Observed government bond yields embed a sovereign default premium. As shown by Miller et al. (2025), higher expected growth lowers the debt-to-GDP ratio on the existing stock of government obligations, reducing the probability of fiscal distress and compressing the default premium. Higher expected growth from AI would have the same effect: the observed yield may therefore rise by less than the default-free rate—or even fall, if the decline in the default premium exceeds the rise in the risk-free rate.

This channel has quantitative bite for countries with high initial debt-to-GDP ratios. If the AI transition raises trend growth by 2 percentage points, the debt-to-GDP ratio on a 30-year horizon falls by roughly 45% relative to the no-AI baseline (holding primary deficits constant). For the United States, with a debt-to-GDP ratio near 120%, this amounts to a reduction to approximately 66%. The default premium on U.S. Treasuries is small in absolute terms, but at the margin, the growth effect could offset 20–40 basis points of the rise in r_f.

6 Historical comparisons

The rare-boom framework was developed by Tsai and Wachter (2016) to study cross-sectional asset pricing in a multisector economy with both positive and negative rare events. Several historical episodes provide precedent for the kind of productivity transformation the AI boom represents.

Table 9 reports other modern sustained growth episodes, expressed as cumulative multipliers on GDP per capita. Historical comparisons are complicated by what one considers the counterfactual, and also by the fact that we have explicitly not considered spillovers from the (at the time of writing) relatively small AI sector.

For the AI rows of Table 9, the GDP column reports the contribution of AI booms alone, with non-AI output held fixed and normal-times growth excluded, as elsewhere in the paper; the historical multipliers include growth from every source, so the comparison understates the AI scenarios. The AI-sector column isolates the sector's own productivity multiplier. With capital fully adjusted, the moderate scenario raises GDP by 8%—a substantial increment, an order of magnitude above task-based estimates, though below the U.S. IT boom over a decade (1.26×). The singularity scenario (2.33×) exceeds the IT boom's decade multiplier. On a 30-year horizon, with booms continuing at $p_* = 0.03$, the moderate scenario's 1.8× is within the range of the industrial revolutions (1.4–2.7× over 50–80 years), the transformative scenario's 3.2× approaches Japan's postwar 4.2× over 25 years, and the singularity scenario's 5.4× lies within the range of the East Asian miracles (4.2–9.0× over 25–30 years).

The U.S. railroad era provides perhaps the closest historical analogue to the current AI buildout. Route mileage rose from approximately 9,000 miles in 1850 to 164,000 in 1890 and continued growing through 1916, peaking at 254,000 miles.[23] The network kept expanding through successive waves of investment. By 2022, the national railroad network

[23] Historical Statistics of the United States, Series Df927; Association of American Railroads.

Table 9: Historical growth episodes and the AI transition

			Multiplier	
Episode	Period	Years	GDP per capita	AI sector
Industrial revolutions				
British Ind. Rev. (1st)	1760–1840	80	1.4	—
U.S. Railroad era	1850–1910	60	2.7	—
2nd Ind. Rev. (U.S.)	1870–1920	50	2.1	—
East Asian growth miracles				
Japan postwar	1945–1970	25	4.2	—
South Korea	1960–1990	30	9.0	—
Taiwan	1960–1990	30	7.3	—
Singapore	1965–1990	25	5.7	—
China	1990–2020	30	5.8	—
Technology booms				
U.S. IT boom	1995–2005	10	1.26	—
AI window scenarios (permanent GDP effects of booms through 2030)				
AI moderate (1 boom)	2024–2030	6	1.08	3.5
AI transformative (2 booms)	2024–2030	6	1.36	12.1
AI singularity (3 booms)	2024–2030	6	2.33	42.1
AI post-window, 30-year horizon (booms continue at $p_* = 0.03$)				
AI moderate	2030–2060	30	1.8	8.6
AI transformative	2030–2060	30	3.2	8.6
AI singularity	2030–2060	30	5.4	8.6

Notes: Historical multipliers are real GDP per capita in 2011 international dollars from the Maddison Project Database 2023 (Bolt and van Zanden, 2025); over the periods shown, the database's series are those of Broadberry et al. (2015) for Britain and Cha et al. (2022) for South Korea. Track mileage from Historical Statistics of the United States, Series Df927. For the AI rows, the GDP column reports the cumulative increase in GDP from AI alone, with non-AI output held at its pre-boom level and normal-times growth excluded, as elsewhere in the paper. For the window scenarios, Period and Years identify when the productivity booms occur; the GDP multipliers measure their permanent effects after capital has fully adjusted (Table 7). On the partial-adjustment paths of Figure 3, the 2030 values are 1.10, 1.35 and 1.84: above the steady state in the moderate scenario, because capital built during the window at the elevated boom probability exceeds the long-run optimum, and below it in the transformative and singularity scenarios, where capital is still catching up with the later booms. The 30-year rows are $s_a(1+g)^{30} + (1-s_a)$, with s_a the scenario's post-window AI output share (Table 7) and $g = p_*(e^{\xi} - 1) \approx 7.4\%$ the expected growth rate of the AI sector. A proportional increase in GDP is the same per capita. The AI-sector column reports the sector's own productivity multiplier ($e^{n\xi}$ for n window booms; $(1+g)^{30}$ post-window); it has no counterpart in the historical episodes.

had contracted to approximately 136,000 route miles, about 46% below its 1916 peak (Bureau of Transportation Statistics, 2024, p. 1-24).

This contraction unfolded over many decades and does not by itself establish that the original infrastructure investment was wasteful. GDP per capita rose 2.7× over the railroad era (Table 9). The original productivity gain was real; the subsequent decline was a negative shock to the sector's z from a competing technology, not a revelation that the gain itself was illusory. For the AI transition, the railroad precedent suggests that transformative booms can involve capital-stock overshooting, but only after long intervals and without negating the underlying productivity gain. The relevant question is whether the productivity boom justifies the investment at the time it is made—not whether some fraction of today's GPU capacity will eventually be stranded, since depreciation at $\delta_a = 0.25$ already implies that current hardware will be largely replaced within four years.

Investment shares provide a second point of comparison. Railroad capital expenditures reached roughly 5–6% of GNP in the peak years of the early 1870s and absorbed about one-fifth of gross domestic fixed capital formation over 1870–1874.[24] Private fixed investment in communication equipment and structures peaked at 1.2% of GDP in 1999–2000, the height of the telecommunications cycle and the largest communications share in the BEA series.[25] The AI shares of Table 2 are 2.5% of GDP in 2026 and 3.3% in 2027—two to three times the telecom peak, and approaching the railroad-era share of output—and, at 18.5% of private fixed investment in 2027, AI capex essentially matches the railroads' share of capital

[24]Computed from railroad gross capital expenditures in Ulmer (1960), Table C-1, and the nominal GNP series of Gallman and Rhode (2020). Because the annual interpolation underlying single-year values is imprecise, the five-year ratio is the more reliable number: cumulative 1870–1874 capital expenditure relative to cumulative GNP is approximately 4%. The early-1880s boom peaked at roughly 3.3–3.9% of GNP. Fishlow (1966) argues that replacement accounting leads Ulmer's series to understate gross investment, so these shares are, if anything, lower bounds.

[25]Bureau of Economic Analysis: private fixed investment in communication equipment ($103B) plus communication structures ($19B) in 2000, relative to GDP of $10.25T; the 1999 share is nearly identical, and no other year since the series began in 1959 comes close. Capital expenditure by publicly traded telecommunications service companies peaked at $121B in 2000 (Doms, 2004), the same share.

formation.

7 Conclusion

The five U.S. technology firms that account for the majority of AI infrastructure investment are scaling capital expenditure at roughly 50% per year, from $159 billion in 2022 to a forecast $784 billion in 2026. This paper embeds this observation in a two-sector model with rare productivity booms to ask what the investment path implies for growth and asset prices.

Calibrating the boom size to match the capital stock built by investment through 2027 implies that the boom raised AI-sector productivity by a factor of roughly 3.5. A two-year window of elevated boom probability generates three scenarios: a moderate outcome (one boom only), a transformative outcome (one further boom), and a singularity scenario (two additional booms). The moderate scenario alone implies a permanent increase in the level of GDP of approximately 8%. By 2030, AI raises GDP relative to the pre-boom economy by approximately 10% in the moderate scenario, 35% in the transformative scenario, and 84% in the singularity scenario.

In our model, the growth does not arise from economy-wide productivity gains. The non-AI sector is not exposed to the boom; these GDP gains hold its output fixed and exclude normal-times growth. What changes is the AI sector's share of the economy, which rises from roughly 3% before the boom to between 10% and 58% depending on the scenario, once capital has fully adjusted. As this share grows, the AI sector's rapid productivity gains increasingly dominate aggregate GDP growth.

With risk aversion of 3, and an EIS of one, long-term expected annual growth reaches approximately 7%, the risk-free rate increases by approximately a third of a percentage point, and the equity premium rises by approximately 3.5 percentage points.

There is a vast range of scenarios, and then substantial long-run uncertainty built on top of those. At its base, however, is a productivity boom that is in investment data, but not yet in productivity data. If the boom fails to materialize, the current buildout will be the largest misallocation of capital in history. On the other hand, those who are certain of that outcome may be making a mistake analogous to that which many forecasters made in 2007, but in reverse. Rare events are hard to imagine until they occur.

References

Acemoglu, D. (2024). The simple macroeconomics of AI. *NBER Working Paper 32487*.

Alphabet Inc. (2023). Alphabet announces first quarter 2023 results. Earnings release, April 25, 2023; Exhibit 99.1 to Form 8-K. Available at https://www.sec.gov/Archives/edgar/data/1652044/000165204423000041/googexhibit991q12023.htm.

Amazon.com, Inc. (2023). 2022 annual report on form 10-K. U.S. Securities and Exchange Commission, fiscal year ended December 31, 2022. Available at https://www.sec.gov/Archives/edgar/data/1018724/000101872423000004/amzn-20221231.htm.

Amazon.com, Inc. (2025). 2024 annual report on form 10-K. U.S. Securities and Exchange Commission, fiscal year ended December 31, 2024. Available at https://www.sec.gov/Archives/edgar/data/1018724/000101872425000004/amzn-20241231.htm.

Amodei, D. (2024). Machines of loving grace: How AI could transform the world for the better. Essay, October 2024. Available at https://darioamodei.com/essay/machines-of-loving-grace.

Amodei, D. (2026). We are near the end of the exponential. Interview by Dwarkesh Patel, *Dwarkesh Podcast*, February 13, 2026. Transcript available at https://www.dwarkesh.com/p/dario-amodei-2.

Babina, T., Fedyk, A., He, A., and Hodson, J. (2024). Artificial intelligence, firm growth, and product innovation. *Journal of Financial Economics*, 151:103745.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.

Bolt, J. and van Zanden, J. L. (2025). Maddison-style estimates of the evolution of the world economy: A new 2023 update. *Journal of Economic Surveys*, 39(2):631–671.

Broadberry, S., Campbell, B. M. S., Klein, A., Overton, M., and van Leeuwen, B. (2015). *British Economic Growth, 1270–1870*. Cambridge University Press.

Bureau of Transportation Statistics (2024). Transportation statistics annual report 2024. Statistical report, U.S. Department of Transportation, Washington, DC. Available at https://doi.org/10.21949/e0kq-gf72.

Cha, M. S., Kim, N. N., Park, K.-J., and Park, Y., editors (2022). *Historical Statistics of Korea*. Studies in Economic History. Springer, Singapore.

Chen, A. Y. (2026). Hedging the singularity. *Working Paper, Federal Reserve Board*. Available at https://arxiv.org/abs/2604.16997.

Cooley, T. F. and Prescott, E. C. (1995). Economic growth and business cycles. In Cooley, T. F., editor, *Frontiers of Business Cycle Research*, pages 1–38. Princeton University Press.

Dasgupta, P. and Stiglitz, J. (1980). Industrial structure and the nature of innovative activity. *Economic Journal*, 90(358):266–293.

Doms, M. (2004). The boom and bust in information technology investment. *Federal Reserve Bank of San Francisco Economic Review*, pages 19–34.

Epstein, L. G. and Zin, S. E. (1989). Substitution, risk aversion, and the temporal behavior of consumption and asset returns: A theoretical framework. *Econometrica*, 57(4):937–969.

Fishlow, A. (1966). Productivity and technological change in the railroad sector, 1840–1910. In Brady, D. S., editor, *Output, Employment, and Productivity in the United States after 1800*, pages 583–646. Columbia University Press for the NBER, New York.

Gallman, R. E. and Rhode, P. W. (2020). *Capital in the Nineteenth Century*. University of Chicago Press, Chicago.

Gomes, J. F., Grotteria, M., and Wachter, J. A. (2019). Cyclical dispersion in expected defaults. *Review of Financial Studies*, 32(4):1275–1308.

Gourio, F. (2012). Disaster risk and business cycles. *American Economic Review*, 102(6):2734–2766.

Grace, K., Stewart, H., Sandkühler, J. F., Thomas, S., Weinstein-Raun, B., and Brauner, J. (2024). Thousands of AI authors on the future of AI. *arXiv preprint arXiv:2401.02843*.

Guo, H. and Wachter, J. A. (2025). "Superstitious" investors. *Review of Asset Pricing Studies*, 15(1):1–45.

Hayashi, F. (1982). Tobin's marginal q and average q: A neoclassical interpretation. *Econometrica*, 50(1):213–224.

Jorgenson, D. W. (1963). Capital theory and investment behavior. *American Economic Review*, 53(2):247–259.

Kording, K. and Marinescu, I. (2025). (Artificial) intelligence saturation and the future of work. Working paper, Brookings Institution, Center on Regulation and Markets. Available at https://www.brookings.edu/articles/artificial-intelligence-saturation-and-the-future-of-work/.

Korinek, A. and Suh, D. (2024). Scenarios for the transition to AGI. *NBER Working Paper 32255*.

Meta Platforms, Inc. (2025). 2024 annual report on form 10-K. U.S. Securities and Exchange

Commission, fiscal year ended December 31, 2024. Available at https://www.sec.gov/Archives/edgar/data/1326801/000132680125000017/meta-20241231.htm.

Microsoft Corporation (2023). Annual report 2023. Annual report for the fiscal year ended June 30, 2023. Available at https://www.microsoft.com/investor/reports/ar23/index.html.

Miller, M., Paron, J. D., and Wachter, J. A. (2025). Sovereign default and the decline in interest rates. *Review of Financial Studies*. Forthcoming.

Musk, E. (2026a). Breaking: Elon Musk's first interview since jury rejected claim against Altman's OpenAI. Interview by Randall Lane, *Forbes*, May 18, 2026. Video, approximately 13:30–15:00.

Musk, E. (2026b). Terawatt of GPUs in space. Interview by Dwarkesh Patel, *Dwarkesh Podcast*, February 5, 2026. Transcript available at https://www.dwarkesh.com/p/elon-musk.

Restrepo, P. (2025). We won't be missed: Work and growth in the AGI world. *NBER Working Paper 34423*.

Space Exploration Technologies Corp. (2026). Form S-1 registration statement. U.S. Securities and Exchange Commission, filed May 20, 2026, accession 0001628280-26-036936. Available at https://www.sec.gov/Archives/edgar/data/1181412/000162828026036936/spaceexplorationtechnologi.htm.

Trammell, P. and Korinek, A. (2023). Economic growth under transformative AI. *NBER Working Paper 31815*.

Tsai, J. and Wachter, J. A. (2016). Rare booms and disasters in a multisector endowment economy. *Review of Financial Studies*, 29(5):1113–1169.

Ulmer, M. J. (1960). *Capital in Transportation, Communications, and Public Utilities: Its Formation and Financing*. Princeton University Press, Princeton.

Van Nieuwerburgh, S. (2026). Financing the AI buildout. *Journal of Economic Perspectives*. Forthcoming; draft of March 20, 2026.

Vanguard Group (2025). AI exuberance: Economic upside, stock market downside. Vanguard Research. Available at https://corporate.vanguard.com/content/corporatesite/us/en/corp/vemo/ai-exuberance-economic-upside-stock-market-downside.html.

Wachter, J. A. (2013). Can time-varying risk of rare disasters explain aggregate stock market volatility? *Journal of Finance*, 68(3):987–1035.

Wachter, J. A. and Kahana, M. J. (2024). A retrieved-context theory of financial decisions. *Quarterly Journal of Economics*, 139(2):1095–1147.

Wachter, J. A. and Zhu, Y. (2025). Learning with rare disasters. *Quantitative Economics*, 16(4):1181–1221.

Yotzov, I., Barrero, J. M., Bloom, N., et al. (2026). Firm data on AI. *NBER Working Paper 34836*.

A Further details on data

A.1 Capital-expenditure data sources

This appendix documents the EDGAR filings underlying the capital-expenditure figures for 2022–2025 in Table 1, specifying the exact line item extracted from each firm's cash-flow statement and the convention used to map filings to calendar years for Microsoft (fiscal year ending June 30) and Oracle (fiscal year ending May 31). Tables A.1 and A.2 report the verbatim figures.

Calendar-year basis. Table 1 is on a calendar-year basis in every column. Amazon, Alphabet, and Meta report on the calendar year, and Table A.1 reports their figures. For Microsoft and Oracle, let fiscal year T be the fiscal year that ends in calendar year T. Capital expenditure for year T is capital expenditure in

$$\text{fiscal year } T \;-\; \text{first six months of fiscal year } T \;+\; \text{first six months of fiscal year } T+1.$$

The fiscal-year figure is from the Form 10-K. The six-month figures are from the Form 10-Q for the second fiscal quarter, whose cash-flow statement covers the first six months of the fiscal year. For Microsoft, the first six months of the fiscal year are July through December, and the construction gives the calendar year, January through December. Oracle's fiscal quarters end in August, November, February, and May, so its filings contain no figure for a period that ends in December. The first six months of Oracle's fiscal year are June through November, and the construction gives the twelve months from December of year $T-1$ through November of year T. The Oracle figure for each year therefore begins and ends one month earlier than the calendar year. Table A.2 reports the filed figures and the construction for both firms, and Figure A.1 shows the construction for 2025.

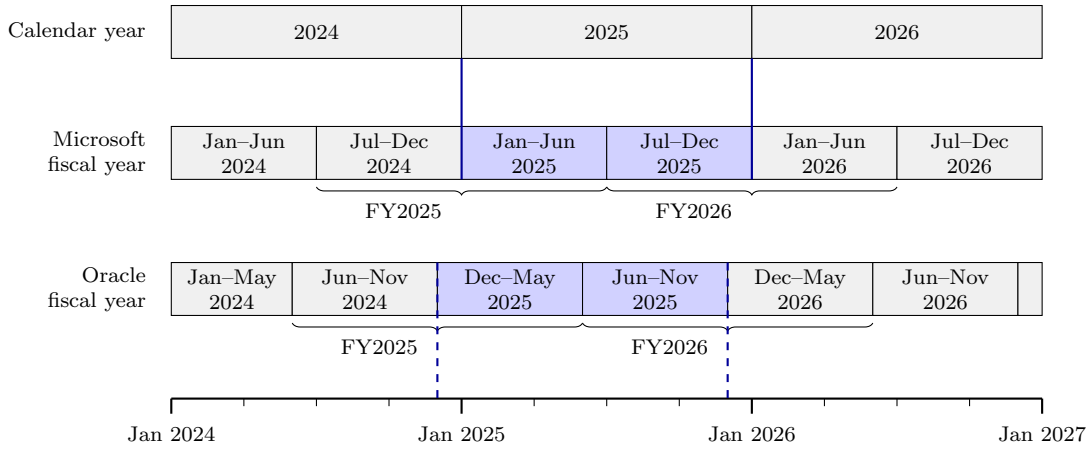


Figure A.1: Fiscal years and the calendar-year construction for Microsoft and Oracle. The shaded half-years make up the 2025 figure in Table 1. For Microsoft, the first shaded half is fiscal 2025 minus its first six months (the FY2025 Form 10-K minus the second-quarter Form 10-Q), and the second is the first six months of fiscal 2026 (the second-quarter Form 10-Q); together they are January–December 2025. For Oracle the same construction gives December 2024–November 2025 (dashed lines), one month earlier than the calendar year.

Rounding. Each entry in Table 1 is rounded to the nearest billion dollars, and each total is the sum of the entries shown. The calibration in Section 3.3 uses these totals. This appendix gives the unrounded figures from the filings, in millions of dollars. The unrounded five-firm totals for 2022–2025 are \$158,612, \$155,220, \$241,056, and \$414,052 million. Rounded to the nearest billion, these are 159, 155, 241, and 414; the totals in Table 1 are 159, 155, 242, and 413.

Timing. Depreciation is annual, and each year's investment is treated as spent at the end of the year. AI capital follows the law of motion (2),

$$K_{a,t+1} = (1 - \delta_a)\, K_{at} + I_{at},$$

where K_{at} is the capital stock at the start of year t and I_{at} is investment during year t, the total for that year in Table 1. Investment of year t does not depreciate during year t and is

in place at the start of year $t + 1$.

The capital stock at the start of 2028 follows from the law of motion and the totals in Table 1. Amounts are in billions of dollars. Before the boom, investment replaces depreciation, so the pre-boom capital stock is the 2022 total divided by the depreciation rate, $K_{a0} = 159/0.25 = 636$. This is the capital stock at the start of 2023. With $\delta_a = 0.25$, the totals for 2023 through 2027 give the capital stock at the start of each following year:

$$\begin{aligned}
&\text{start of 2024:} \quad 0.75 \times 636 + 155 = 632,\\
&\text{start of 2025:} \quad 0.75 \times 632 + 242 = 716,\\
&\text{start of 2026:} \quad 0.75 \times 716 + 413 = 950,\\
&\text{start of 2027:} \quad 0.75 \times 950 + 784 = 1{,}496.5,\\
&\text{start of 2028:} \quad 0.75 \times 1{,}496.5 + 1{,}090 = 2{,}212.375.
\end{aligned}$$

The capital stock at the start of 2028 is $2{,}212.375/636 \approx 3.48$ times the pre-boom capital stock. Section 3.3 calibrates the boom size to this ratio.

Table A.1: Capital expenditure of calendar-year filers: cash-flow-statement values from EDGAR ($ millions)

	2022	2023	2024	2025
Amazon: "Purchases of property and equipment"†	63,645	52,729	82,999	131,819
less: proceeds from P&E sales and incentives	5,324	4,596	5,341	3,499
equals: net cash purchases of P&E	58,321	48,133	77,658	128,320
Alphabet: "Purchases of property and equipment"†	31,485	32,251	52,535	91,447
Meta: "Purchases of property and equipment"	31,186	27,045	37,256	69,691
plus: principal payments on finance leases	850	1,058	1,969	2,524
equals: capex per management definition†	32,036	28,103	39,225	72,215

Notes: Figures are taken verbatim from the Consolidated Statements of Cash Flows in each firm's Form 10-K; rows labeled *equals* are computed from the two rows above them. The column heading is the calendar year, which is the fiscal year of all three firms. † The entry in Table 1 is this row, rounded to the nearest billion dollars.

Table A.2: Capital expenditure of Microsoft and Oracle: filed values and calendar-year construction ($ millions)

Year T	2022	2023	2024	2025	2026
Microsoft: "Additions to property and equipment"; fiscal year T is July $T-1$–June T					
(a) Fiscal year T	23,886	28,107	44,477	64,551	115,948
(b) July–December $T-1$	11,675	12,557	19,652	30,727	49,270
(c) July–December T	12,557	19,652	30,727	49,270	—
(a) − (b): January–June T	12,211	15,550	24,825	33,824	66,678
(a) − (b) + (c): January–December $T^{\dagger}$	24,768	35,202	55,552	83,094	—
Oracle: "Capital expenditures"; fiscal year T is June $T-1$–May T					
(a) Fiscal year T	4,511	8,695	6,866	21,215	55,663
(b) June–November $T-1$	1,987	4,154	2,394	6,273	20,535
(c) June–November T	4,154	2,394	6,273	20,535	—
(a) − (b): December $T-1$–May T	2,524	4,541	4,472	14,942	35,128
(a) − (b) + (c): December $T-1$–November $T^{\dagger}$	6,678	6,935	10,745	35,477	—

Notes: Rows (a), (b), and (c) are taken verbatim from the statements of cash flows. Row (a) is from the Form 10-K. Row (b) is the first six months of fiscal year T and row (c) is the first six months of fiscal year $T+1$; both are from the Form 10-Q for the second fiscal quarter. The 2026 column reports the figures filed as of September 29, 2026; a dash indicates that the Form 10-Q for the second quarter of fiscal 2027 has not been filed. Oracle's Form 10-Q for the first quarter of fiscal 2027 reports 28,499 for June–August 2026, so that December 2025–August 2026 is $35{,}128 + 28{,}499 = 63{,}627$. Appendix A.2 describes the 2026 entries in Table 1. † The 2022–2025 entries in Table 1 are this row, rounded to the nearest billion dollars.

Amazon (CIK 0001018724). The relevant cash-flow line is labeled *Purchases of property and equipment*. A separate line (*Proceeds from property and equipment sales and incentives*) records small recoveries which we report on a second row of Table A.1 for completeness. Equipment acquired under finance leases is reported as a non-cash supplemental disclosure and is excluded from the cash capex line. Sources: FY2022 figures from the FY2024 10-K (accession 0001018724-25-000004, filed February 7, 2025); FY2023–FY2025 figures from the FY2025 10-K (accession 0001018724-26-000004, filed February 6, 2026). The 2026 management forecast of "about $200 billion" was first disclosed in the Q4 2025 earnings release (Form 8-K, February 5, 2026, Exhibit 99.1).

Alphabet (CIK 0001652044). The cash-flow line is labeled *Purchases of property and equipment*. Alphabet does not report finance-lease additions in this line and does not net proceeds from sales against gross purchases at this position. Sources: FY2022 figures from the FY2024 10-K (accession 0001652044-25-000014, filed February 5, 2025); FY2023–FY2025 figures from the FY2025 10-K (accession 0001652044-26-000018, filed February 5, 2026). The 2026 guidance of $175–$185 billion was disclosed in the Q4 2025 earnings release (Form 8-K, February 4, 2026, Exhibit 99.1, attributed to CEO Sundar Pichai). Prospectus supplements (Form 424B5) filed June 2 and June 4, 2026, and the Form 8-K of June 4, 2026 (Exhibit 99.1), state a range of $180–$190 billion. The 2026 value in Table 1 is the midpoint of the later range, $185 billion (Appendix A.2).

Microsoft (CIK 0000789019). The cash-flow line is labeled *Additions to property and equipment*. Assets acquired under finance leases are reported as a non-cash supplemental disclosure, and principal payments on finance leases are reported as financing cash flows (FY2026 10-K, Note 13). Neither is part of this line. Microsoft's fiscal year ends June 30, so the figure reported for "FY2022" covers July 2021–June 2022, "FY2023" covers July 2022–

June 2023, and so on. The figures in Table 1 of the main text are calendar-year figures, constructed in Table A.2: the "2022" column reports January–June 2022 from FY2022 plus July–December 2022 from FY2023, $(23{,}886 - 11{,}675) + 12{,}557 = 24{,}768$ ($ millions), and similarly for later years. Sources: FY2022 figures from the FY2024 10-K (accession 0000950170-24-087843, filed July 30, 2024); FY2023–FY2025 figures from the FY2025 10-K (accession 0000950170-25-100235, filed July 30, 2025); FY2026 figures from the FY2026 10-K (accession 0001193125-26-323660, filed July 29, 2026). Six-month figures are from the 10-Q for the quarter ended December 31 of each year: 2021 (accession 0001564590-22-002324, filed January 25, 2022); 2022 (accession 0001564590-23-000733, filed January 24, 2023); 2023 (accession 0000950170-24-008814, filed January 30, 2024); 2024 (accession 0000950170-25-010491, filed January 29, 2025); 2025 (accession 0001193125-26-027207, filed January 28, 2026). Microsoft does not issue annual capex guidance in any SEC filing (see Microsoft's standard language in each 8-K Exhibit 99.1: "Microsoft will provide forward-looking guidance in connection with this quarterly earnings announcement on its earnings conference call and webcast"). A presentation furnished with a Form 8-K on September 2, 2026 (Item 7.01) restates the outlook that Microsoft gave on July 29, 2026 for the quarter ending September 30, 2026, including capital expenditures. We do not use it as a source for Table 1. The 2026 value in Table 1 is the filed figure for January–June 2026 plus an extrapolation for July–December 2026 (Appendix A.2).

Meta (CIK 0001326801). The cash-flow line is labeled *Purchases of property and equipment*. Meta separately reports *Principal payments on finance leases* in the financing-activities section, and its management commentary defines "capital expenditures" to be the sum of the two—i.e., investing-activities P&E purchases plus financing-activities lease principal. We report both the cash-flow-statement value and the management-definition total in Table A.1. Table 1 uses the management-definition total in every year. Sources: FY2022 figures from

the FY2024 10-K (accession 0001326801-25-000017, filed January 30, 2025); FY2023–FY2025 figures from the FY2025 10-K (accession 0001628280-26-003942, filed January 29, 2026). The 2026 guidance of $130–$145 billion (narrowed from $125–$145 billion in the Q1 2026 earnings release) was disclosed in the Q2 2026 earnings release (Form 8-K, July 29, 2026, Exhibit 99.1) and again in the Q2 2026 10-Q MD&A.

Oracle (CIK 0001341439). The cash-flow line is labeled *Capital expenditures*. Oracle's fiscal year ends May 31, so the figure reported for "FY2022" covers June 2021–May 2022, and so on. The figures in Table 1 of the main text cover the twelve months from December of the preceding year through November, constructed in Table A.2: the "2022" column reports December 2021–May 2022 from FY2022 plus June–November 2022 from FY2023, $(4{,}511 - 1{,}987) + 4{,}154 = 6{,}678$ ($ millions), and similarly for later years. Oracle's 10-Q for the quarter ended November 30 reports capital expenditures for the trailing four quarters in the free cash flow table of the MD&A. These equal the figures constructed in Table A.2. Sources: FY2022 figures from the FY2024 10-K (accession 0000950170-24-075605, filed June 20, 2024); FY2023–FY2025 figures from the FY2025 10-K (accession 0000950170-25-087926, filed June 18, 2025); FY2026 figures from the FY2026 10-K (accession 0001193125-26-277521, filed June 22, 2026). Six-month figures are from the 10-Q for the quarter ended November 30 of each year: 2021 (accession 0001564590-21-060022, filed December 10, 2021); 2022 (accession 0001564590-22-039278, filed December 13, 2022); 2023 (accession 0000950170-23-069682, filed December 12, 2023); 2024 (accession 0000950170-24-134973, filed December 10, 2024); 2025 (accession 0001193125-25-315925, filed December 11, 2025). The June–August 2026 figure is from the 10-Q for the quarter ended August 31, 2026 (accession 0001193125-26-389274, filed September 11, 2026). Oracle disclosed an FY2026 (June 2025–May 2026) capex target of $50 billion in the Q3 FY2026 earnings release (Form 8-K, March 10, 2026, Exhibit 99.1). FY2026 capital expenditures were $55.7 billion (Ta-

ble A.2). Two reporting peculiarities affect interpretation. First, Oracle's reported capex excludes GPUs that customers buy and supply to Oracle. Equipment that Oracle buys with customer prepayments is in the reported line; the prepayments are operating cash inflows ($4,592 million in fiscal 2026). The Q3 FY2026 release states that "most of the equipment needed is either funded upfront via customer prepayments so Oracle can purchase the GPUs, or the customer buys the GPUs and supplies them to Oracle." Section 2 treats the reported figure as the relevant magnitude for the calibration. Second, Oracle delivers data-center capacity to Amazon, Alphabet, and Microsoft under its MultiCloud partnership, raising a potential double-count at the data-center-shell layer (not at the server/GPU layer, which sits in each hyperscaler's separate capex).

Non-GAAP nature of the capex concept. The Consolidated Statement of Cash Flows itself is mandated by GAAP (ASC 230), and a line near "purchases of property and equipment" is required in the investing-activities section. But "capital expenditure" is a managerial concept that firms use in MD&A and earnings guidance, not a GAAP-defined line item. Firms accordingly differ in the label they use, in whether proceeds from sales are netted against gross purchases, in whether finance-lease additions (typically non-cash and disclosed elsewhere) are included, and in whether and how forward guidance is delivered. Table 1 of the main text reflects these conventions firm by firm: the lines for Amazon, Alphabet, Microsoft, and Oracle exclude finance leases, Meta's management-disclosed capex adds finance-lease principal from the financing-activities section to investing-activities P&E purchases, and Oracle reports the gross figure under the label "Capital expenditures." These conventions are documented above and in Tables A.1 and A.2.

A.2 Capital-expenditure forecasts for 2026 and bottom-up estimates for 2027

This appendix documents the firm-by-firm source materials underlying the 2026 and 2027 columns of Table 1. It uses filings available on EDGAR through September 29, 2026. Figures that appear only in earnings-call commentary are not used.

For Amazon, Alphabet, and Meta, the 2026 entry is a figure that management states in a filed document: a point figure for Amazon and the midpoint of a range for Alphabet and Meta. For Microsoft and Oracle, the 2026 entry is the sum of the capital expenditures reported to date and an extrapolation for the remaining months.

Every 2026 entry is on the calendar-year basis of Table 1 (Appendix A.1). The entries for Amazon, Alphabet, Meta, and Microsoft cover January–December 2026. The entry for Oracle covers December 2025–November 2026. Oracle's fiscal quarters end in August, November, February, and May, so its filings contain no figure for a period that ends in December.

Table A.3 reports, in millions of dollars, the capital expenditures reported to date and the full-year figures. The entries in Table 1 are the full-year point figures rounded to the nearest billion dollars.

For 2027, no firm has issued formal annual guidance. Each 2027 estimate is anchored to multi-year commitments already in filings—leases not yet commenced, contracted revenue with capex prerequisites, and signed multi-year customer agreements—together with firm-specific forward-looking statements about contracted demand and capacity in 10-Qs and 8-Ks.

Table A.3: Capital expenditure in 2026: amounts to date and full-year figures ($ millions)

	Actual	Full year			Table 1
	to date	Low	Point	High	($ billions)
Amazon	98,411		200,000		200
Alphabet	80,598	180,000	185,000	190,000	185
Microsoft	66,678	158,775	168,857	178,939	169
Meta	50,918	130,000	137,500	145,000	138
Oracle	63,627	86,123	92,126	102,189	92
Total			783,483		784

Notes: "Actual to date" is capital expenditure in January–June 2026 (Oracle: December 2025–August 2026). "Full year" is January–December 2026 (Oracle: December 2025–November 2026). For Amazon, Alphabet, and Meta, the full-year figures are stated by management, and the point figure for Alphabet and Meta is the midpoint of the stated range. For Microsoft and Oracle, the full-year figures are the actual to date plus an extrapolation for the remaining months. The line item for each firm is the one documented in Appendix A.1; Meta's figures include principal payments on finance leases. The last column is the point figure rounded to the nearest billion dollars; its total is the sum of the rounded entries.

2026 forecasts (five-firm aggregate: $784B)

Amazon: $200B (point). The Q4 2025 earnings release (Form 8-K, February 5, 2026, Exhibit 99.1; accession 0001018724-26-000002) states verbatim:

> *"With such strong demand for our existing offerings and seminal opportunities like AI, chips, robotics, and low earth orbit satellites, we expect to invest about $200 billion in capital expenditures across Amazon in 2026, and anticipate strong long-term return on invested capital."*

The statement is attributed to Andy Jassy, President and CEO. The 2025 letter to shareholders (Form 8-K, April 9, 2026, Exhibit 99.1; accession 0001104659-26-041034) gives approximately $200 billion of capital expenditures in 2026. No later filing gives a figure. The earnings releases for the first and second quarters of 2026 (Forms 8-K, April 29 and July 30, 2026) give no figure for capital expenditures in 2026. The release of February 5, 2026 does not say whether the $200 billion is before or after proceeds from property and equipment

sales and incentives.

Purchases of property and equipment, the line used in Table 1, were $98.411B in January–June 2026 and $57.202B in January–June 2025. Proceeds from property and equipment sales and incentives were $2.101B in January–June 2026. The figures are from the 10-Q for the second quarter (filed July 31, 2026; accession 0001018724-26-000026).

Alphabet: $180–$190B range; $185B midpoint. The prospectus supplement for Alphabet's offering of Class A and Class C stock (Form 424B5, filed June 4, 2026; accession 0001193125-26-256375) states, under "Use of Proceeds":

> *"We expect that capital expenditures in 2026 will be in the range of $180 billion to $190 billion, and that 2027 capital expenditures will significantly increase compared to 2026."*

Five other prospectus supplements (Form 424B5) that Alphabet filed on June 2 and June 4, 2026 contain the same sentence. The press release furnished with the Form 8-K of June 4, 2026 (Exhibit 99.1; accession 0001193125-26-257724) gives the same range. The earlier range of $175 to $185 billion is part of the chief executive's statement in the Q4 2025 earnings release (Form 8-K, February 4, 2026, Exhibit 99.1; accession 0001652044-26-000012). The earnings releases for the first and second quarters of 2026 give no figure. The 10-Qs for those quarters state: "In 2026, we expect to significantly increase, relative to 2025, our investment in our technical infrastructure, including servers and network equipment and data centers."

Purchases of property and equipment were $80.598B in January–June 2026 and $39.643B in January–June 2025 (10-Q filed July 23, 2026; accession 0001652044-26-000071).

Microsoft: $159–$179B range; $169B midpoint (January–June actual plus extrapolation). Microsoft does not issue annual capital-expenditure guidance in its earnings

releases, its 10-Qs, or its 10-K. Each earnings release (Form 8-K, Exhibit 99.1), including the release of July 29, 2026, contains the same boilerplate:

> *"Microsoft will provide forward-looking guidance in connection with this quarterly earnings announcement on its earnings conference call and webcast."*

Forward dollar figures are delivered verbally on quarterly calls. A presentation furnished with the Form 8-K of September 2, 2026 (Item 7.01, Exhibit 99.1; accession 0001193125-26-380280) restates the outlook that Microsoft gave on July 29, 2026 for the quarter ending September 30, 2026. It lists capital expenditures for that quarter as "Expected to be over $50 billion including the impact of the useful life update" and does not define capital expenditures. We do not use it as a source for Table 1.

Calendar 2026 spans the second half of FY2026 (January–June 2026) plus the first half of FY2027 (July–December 2026). January–June 2026 is reported. Additions to property and equipment were $115.948B in FY2026 (10-K filed July 29, 2026; accession 0001193125-26-323660) and $49.270B in the six months ended December 31, 2025 (10-Q filed January 28, 2026; accession 0001193125-26-027207). January–June 2026 is the difference, $115.948B − $49.270B = $66.678B. The same construction gives $33.824B for January–June 2025. The quarter ended June 30, 2026 accounts for $35.802B of the $66.678B (earnings release, Form 8-K, July 29, 2026, Exhibit 99.1; accession 0001193125-26-323632).

July–December 2026 is an extrapolation from the June quarter. We assume that additions to property and equipment grow at a rate g per quarter of between 18% and 34%, so that July–December is $(1+g)+(1+g)^2$ times the June quarter. For comparison, additions to property and equipment in the four quarters of FY2026 were $19.4B, $29.9B, $30.9B, and $35.8B (earnings releases for the four quarters), growth of 54%, 3%, and 16% from one quarter to the next. July–December 2026 is $35.802B $\times$ 2.572 = $92.1B at 18% and $35.802B $\times$ 3.136 = $112.3B at 34%. Calendar 2026 is $158.8B at the lower end and $178.9B

at the upper end. The midpoint is $168.9B.

Additions to property and equipment is an investing cash flow and does not include finance leases. The FY2026 10-K (Note 13) reports assets acquired under finance leases as obtained in exchange for lease obligations ($24.6B in FY2026) and reports cash paid on finance leases in operating and financing cash flows ($2.5B and $3.1B).

Meta: $130–$145B range; $137.5B midpoint. The Q2 2026 earnings release (Form 8-K, July 29, 2026, Exhibit 99.1; accession 0001628280-26-050596) states verbatim:

> *"We anticipate 2026 capital expenditures, including principal payments on finance leases, to be in the range of $130–145 billion, narrowed from our prior outlook of $125–145 billion."*

The 10-Q for the second quarter (filed July 30, 2026; accession 0001628280-26-050705) gives the same range. Table 1 rounds the midpoint to $138B. Meta's disclosed figure is the "capex plus finance-lease principal" composite documented in Table A.1. Table 1 uses this composite for Meta in every year. January–June 2026 actuals were $50.918B ($49.113B in purchases of property and equipment plus $1.805B in finance-lease principal; same 10-Q), and January–June 2025 actuals were $30.704B. Meta funded the buildout in part through $29.9B of net proceeds from fixed-rate senior unsecured notes issued in November 2025 (FY2025 10-K).

Oracle: $92B; $86–$102B range (December–August actual plus extrapolation). Oracle's fiscal year ends May 31, so calendar 2026 is approximated by the third and fourth quarters of fiscal 2026 plus the first and second quarters of fiscal 2027 (December 2025 through November 2026, one month offset from calendar 2026 proper). The first three of these quarters are reported. Capital expenditures were $55.663B in fiscal 2026 (10-K filed June 22, 2026; accession 0001193125-26-277521) and $20.535B in the six months ended November 30,

2025 (10-Q filed December 11, 2025; accession 0001193125-25-315925). December 2025–May 2026 is the difference, \$55.663B − \$20.535B = \$35.128B. Capital expenditures in the quarter ended August 31, 2026 were \$28.499B (10-Q filed September 11, 2026; accession 0001193125-26-389274). December 2025–August 2026 is \$35.128B + \$28.499B = \$63.627B.

Oracle has not stated a dollar figure for capital expenditures in fiscal 2027 in a filed document. The 10-Q for the quarter ended August 31, 2026 states that Oracle expects capital expenditures in fiscal 2027 to be higher than in fiscal 2026. For September–November 2026 we set capital expenditures equal to those of the August quarter, \$28.499B. The entry is \$63.627B + \$28.499B = \$92.126B.

The range uses the quarterly capital expenditures in the first-quarter fiscal 2027 earnings release (Form 8-K, September 10, 2026, Exhibit 99.1; accession 0001193125-26-387905): \$8.502B, \$12.033B, \$18.635B, and \$16.493B in the four quarters of fiscal 2026, and \$28.499B in the first quarter of fiscal 2027. The lower end sets September–November 2026 equal to the average of the two most recent quarters, (\$16.493B + \$28.499B)/2 = \$22.496B, which gives \$86.1B for the twelve months. The upper end lets capital expenditures grow from the August quarter at the compound quarterly growth rate between the quarter ended August 31, 2025 and the quarter ended August 31, 2026, $(28.499/8.502)^{1/4} - 1 = 35\%$, to \$38.6B, which gives \$102.2B.

Oracle's \$50B financing program announced in February 2026 (senior notes, mandatory-convertible preferred, and an at-the-market equity facility, disclosed in the corresponding 8-K exhibits) was sized specifically to fund the FY2027 buildout; \$30B was raised within days of announcement.

Oracle's reported capex line excludes GPUs that customers buy and supply to Oracle under its large-scale AI contracts. Equipment that Oracle buys with customer prepayments is in the reported line; the prepayments appear as operating cash inflows (\$4.6B in fiscal 2026 and \$11.4B in the quarter ended August 31, 2026). The Q3 FY2026 release states verbatim:

> *"Most of the increase in RPO in Q3 related to large scale AI contracts where Oracle does not expect to have to raise any incremental funds to support these contracts as most of the equipment needed is either funded upfront via customer prepayments so Oracle can purchase the GPUs, or the customer buys the GPUs and supplies them to Oracle."*

The Q4 FY2026 release (Form 8-K, June 10, 2026, Exhibit 99.1; accession 0001193125-26-265848) puts the prepaid and customer-supplied hardware portions of these contracts at $75B in total, without dividing the two. The reported capex line therefore understates the compute deployed through Oracle data centers by the customer-supplied portion, which Oracle does not disclose separately. Table 1 uses the reported figure.

2027 bottom-up estimates (five-firm aggregate: $1,090B)

No firm has issued formal full-year 2027 capital-expenditure guidance in a filed document. The 2027 (est.) column of Table 1 reports bottom-up estimates that, for each firm, are bounded below by mechanical commitments visible in filings and bounded above by recent quarterly run-rates. The estimates were made before the filings of June through September 2026 and are not revised.

Amazon: $285B. Amazon's cash capex grew $53B (FY2023) → $83B (FY2024, +57%) → $132B (FY2025, +59%) → $200B (FY2026 guided, +52%). The $285B 2027 estimate (range $280–$310B) is supported by four convergent forward anchors in the Q1 2026 earnings release (Form 8-K, April 29, 2026), Exhibit 99.1:

- An OpenAI commitment to consume approximately 2 GW of Trainium capacity through AWS infrastructure "[which] begins ramping in 2027."

- An Anthropic commitment "to secure up to five gigawatts (GW) of current and future generations of Amazon's Trainium chips to train and power their advanced AI models."
- Trainium4 production launch in 2027, "with 6 times the FP4 compute performance, 4 times more memory bandwidth, and 2 times more high memory bandwidth capacity than Trainium3."
- Deployment of more than one million NVIDIA GPUs starting in 2026 (Q4 2025 release), with revenue recognition extending through 2027 and beyond.

At a cost of $40B per gigawatt, the 2 GW contracted OpenAI Trainium build alone implies an $80B envelope, with the Anthropic 5 GW ramp, Trainium4 production capacity, and NVIDIA GPU deployment program layered on top.

Alphabet: $255B. Alphabet's cash capex grew $32B (FY2023) → $53B (FY2024, +63%) → $91B (FY2025, +74%) → $185B (FY2026 midpoint, +102%). January–June 2026 actual capex of $80.6B is 44% of the $185B 2026 midpoint. Alphabet has not stated a dollar figure for 2027; the prospectus supplements of June 2026 state that 2027 capital expenditures "will significantly increase compared to 2026." The $255B 2027 point estimate corresponds to roughly +38% growth over the 2026 midpoint and is anchored by four multi-year commitments disclosed in filings:

- Multi-gigawatt TPU hardware supply agreements with customers, disclosed in the Q1 2026 10-Q MD&A: "Google Cloud has entered into a limited number of agreements to supply multiple gigawatts of TPU hardware to customers who require or provide on-premises infrastructure for specialized, high-scale workloads.... We expect to begin recognizing revenues from these agreements later in 2026, with the significant majority to be recognized in 2027." The host disclosure is the $460B Google Cloud backlog cited in the Q1 2026 8-K Exhibit 99.1.

- Future lease payments on leases not yet commenced, primarily for data centers, totaling $75.6B at March 31, 2026 (Q1 2026 10-Q, Note 4), up $17B in a single quarter from $58.5B at December 31, 2025. These leases commence between 2026 and 2031 with non-cancelable terms primarily between one and 25 years.
- A $33.3B forward-backstop framework, disclosed for the first time in the Q1 2026 10-Q, to "support the build-out of data center and energy supply infrastructure," against which a $15.3B specific-provider backstop was finalized in April 2026. Combined with $37.4B of pre-existing committed backstops (financial guarantees plus credit derivatives), total off-balance-sheet support for third-party infrastructure exceeds $70B at quarter-end.
- A $9.9B power-purchase agreement covering 2027–2047, executed January 2026, expected to be accounted for as a lease (FY2025 10-K).

Total purchase commitments at March 31, 2026, stood at $332.4B, "primarily for technical infrastructure" (Q1 2026 10-Q).

Microsoft: $250B. Filed additions to property and equipment totaled $115.948B in FY2026, a +80% increase versus FY2025. The most recent reported quarter, Q4 FY2026 (April–June 2026), was $35.802B alone, a +110% year-over-year increase. Calendar-2026 capex of $169B (midpoint of $159–$179B) is the January–June 2026 actual plus an extrapolation of the June quarter through December 2026. The $250B 2027 figure is bounded below by five mechanical commitments visible in the filings:

- Leases primarily for data centers that had not yet commenced stepped from $92.7B (June 30, 2025) to $155.1B (December 31, 2025) to $196.6B (March 31, 2026) to $329.1B (June 30, 2026), with commencement spanning fiscal years 2027–2033. The

FY2026 10-K states that some of these arrangements are subject to contractual conditions.

- Servers, network equipment, and software on the balance sheet grew from $132.8B (June 30, 2025) to $190.9B (March 31, 2026)—a +44% gross-stock increase in nine months reflecting the AI-compute layer.
- Accounts payable for purchases of property and equipment rose from $6.9B (June 30, 2025) to $22.6B (March 31, 2026), a $15.7B in-flight invoice pipeline implying near-term cash conversion.
- Commercial remaining performance obligation reached $627B at March 31, 2026, up +99% year-over-year—equivalent to roughly three years of Microsoft Cloud revenue contracted forward. AI-business annual revenue run-rate reached $37B, up +123% year-over-year.
- Microsoft's strategic partnership with OpenAI gives Microsoft "a right of first refusal on OpenAI's new capacity needs" (FY2025 10-K), tying Azure capex to the largest single source of AI compute demand.

The lease-commencement pipeline and the asset-stock trajectory together imply a step-up in calendar-2027 cash capex relative to calendar 2026.

Meta: $195B. Meta's reported capex (including principal payments on finance leases) grew $28B (FY2023) → $39B (FY2024, +40%) → $72B (FY2025, +84%) → $137.5B (FY2026 midpoint, +90%). The $195B 2027 estimate is supported by four firm-specific indicators:

- The Q1 2026 earnings release attributes part of the $10B upward revision to the 2026 capex guidance range to "higher component pricing this year and, to a lesser extent,

additional data center costs to support *future year capacity*" (Form 8-K, April 29, 2026, Exhibit 99.1). "Future year capacity" is the in-filing 2027 anchor: a portion of the 2026 raise is explicitly pre-funding 2027 capacity.

- The Q1 2026 10-Q MD&A states: "We have made significant investments in AI initiatives . . . and expect to continue to increase these investments. . . . [W]e have significantly increased our infrastructure investments in connection with our AI initiatives . . . and expect our investments to continue to increase."
- January–June 2026 actual capex of $50.9B (+66% versus January–June 2025) leaves $86.6B for July–December at the $137.5B full-year midpoint, an average of $43.3B per quarter. That quarterly rate annualizes to $173B for 2027 before any further growth.
- In November 2025, Meta raised $29.9B of net proceeds from fixed-rate senior unsecured notes (FY2025 10-K), funding the multi-year AI buildout.

A multi-year, multi-billion-dollar CoreWeave capacity contract was reported in March 2026; under ASC 606 service-contract accounting it does not enter Meta's capex line, and the corresponding capex is reported separately in Appendix A.3.

Oracle: $105B. Oracle's 2027 figure is built from a different bridge. The Q3 FY2026 earnings release (March 10, 2026, Form 8-K) raised FY2027 total revenue guidance to $90B, an increase from approximately $67B in FY2026:

> *"For fiscal year 2027, we are raising total revenue guidance to $90 billion."*

The Q1 FY2026 release (Form 8-K, September 9, 2025) disclosed a five-year Oracle Cloud Infrastructure revenue trajectory—FY2026 $18B, FY2027 $32B, FY2028 $73B, FY2029 $114B, FY2030 $144B—requiring a major capital-expenditure step-up beginning in FY2027. Holding the guided FY2026 capex-to-revenue ratio ($50B/$67B = 0.75; Q3 FY2026 release)

constant against the $90B FY2027 revenue guidance gives FY2027 capex of approximately $67B. This is best read as a floor: OCI capital intensity is rising steeply—FY2028 OCI revenue of $73B (from $32B in FY2027) is a 2.3× step—and the February 2026 $50B financing program was sized to fund the FY2027 buildout, so a reasonable FY2027 range is $80–$120B. Applying the same capex-to-revenue ratio to projected FY2028 total revenue of roughly $115–$125B (FY2028 OCI plus continued growth in non-OCI lines) gives FY2028 capex of approximately $85–$95B. Calendar 2027 (December 2026–November 2027) spans the second half of FY2027 and the first half of FY2028, both within the ramp; splitting the calendar year across the two fiscal years gives a point estimate of approximately $105B, with a range of $85–$135B.

The trajectory of recent actuals supports the ramp: FY2026 capex was $55.66B (+162% versus FY2025), above the $50B guidance, and Q1 FY2027 (June–August 2026) alone was $28.50B (+235% year-over-year). Remaining performance obligation reached $553B at Q3 FY2026, up +325% year-over-year, with Q3 alone adding $29B—substantially representing large-scale AI contracts. The $105B figure excludes GPUs that customers buy and supply to Oracle, as described in the 2026 section above.

Reconciliation. The five-firm 2026 forecasts sum to $200 + $185 + $169 + $138 + $92 = $784B, matching the 2026 column of Table 1. The five-firm 2027 bottom-up estimates sum to $285 + $255 + $250 + $195 + $105 = $1,090B, the value reported in the "2027 (est.)" column of Table 1. Both totals reflect the paper's five-firm scope. Including the sixth hyperscaler (xAI, the AI segment of SpaceX following the February 2026 acquisition), at an estimated $40B in 2026 (anchored to the $7.7B Q1 2026 AI-segment capex disclosed in SpaceX's S-1) and $60B in 2027, raises the totals to $824B and $1,150B, respectively; the xAI evidence is detailed in Appendix A.3. Bottom-up reconstructions that incorporate U.S.-domiciled neocloud operators place 2027 U.S. AI infrastructure capex at approximately $1,220B. The

Tier 1 neocloud increment (CoreWeave, Crusoe, IREN, Lambda) is fully incremental to the five-firm total under ASC 606 service-contract accounting; two partial-overlap names (Applied Digital and Hut 8) are excluded because of potential double-count with hyperscaler capex (Appendix A.3). The $1,090B five-firm figure in Table 1 is therefore a conservative measure of the relevant total.

A.3 Capital expenditure outside the five-firm scope

The five firms in Table 1 account for the bulk of U.S. AI infrastructure capex through 2025 but not the entire universe. This subsection documents two additional categories of spending omitted from the main-text table: a sixth hyperscaler (xAI) and four "neocloud" service providers whose revenue is recognized under ASC 606 service-contract accounting (CoreWeave, Crusoe, IREN, Lambda). A third category—"Tier 2" partial-overlap firms (Applied Digital, Hut 8), whose revenue includes ASC 842 operating-lease arrangements with the hyperscalers—is addressed at the end of this appendix.

Why the Tier 1 neoclouds are fully incremental to the five-firm aggregate. CoreWeave's FY2025 10-K (Note 2, Revenue Recognition) states verbatim:

> *"[The Company's] arrangements do not meet the definition of a lease under ASC 842 and are accounted for as service contracts under ASC 606."*

In consequence, a hyperscaler purchasing AI compute from CoreWeave (or any Tier 1 peer with the same accounting structure) records the payment as an operating expense (cost of revenue or, depending on use, research and development), not as a finance-lease right-of-use asset. CoreWeave's underlying property and equipment remains on CoreWeave's balance sheet. Adding CoreWeave's reported capex to the five-firm aggregate therefore introduces

no double-count.[26] xAI is in a different position: it owns and operates its own data centers, is not a service provider to the hyperscalers, and is treated as a sixth hyperscaler that is simply not publicly traded. In all five cases below, the spending is incremental to Table 1.

Historical capex by firm, 2022–2025. Table A.4 reports annual capital expenditure for the five additional firms. Two (CoreWeave and IREN) file with the SEC directly and have audited cash-flow disclosures. xAI, originally a private standalone firm, was acquired by SpaceX in February 2026 and is reported in SpaceX's May 2026 S-1 as the AI segment of the combined entity (Space Exploration Technologies Corp., 2026); the 2023–2025 figures shown for xAI are the audited AI-segment capital-expenditure line from the Consolidated Statements of Cash Flows. Crusoe and Lambda remain private with no audited filings, and defensibility for those two rows is correspondingly weaker. Before 2024, all five firms were either pre-AI in business model (IREN was a bitcoin miner; Crusoe was a flare-gas/crypto firm), pre-founding (xAI was founded in mid-2023), or very small (CoreWeave's 2023 capex was $3.4B economic).

Comparison to the five-firm aggregate. The 2022–2023 contribution from this expanded universe is negligible relative to the $159B and $155B five-firm aggregates. In 2024, the $17B expanded-universe contribution is approximately 7% of the five-firm $242B; in 2025, the $41B contribution is approximately 10% of the five-firm $413B. These percentages mix measurement bases—CoreWeave's economic capex includes OEM-financed additions, while the five-firm aggregate follows the cash-flow conventions of Table 1—and are therefore approximate. The expanded universe grows substantially in 2026 (forecast contribution ap-

[26] An alternative model—under which the hyperscaler would consolidate the leased compute as its own asset—would require the lease to convey control of an identified asset under ASC 842. The 10-K language above rules out this classification. IREN explicitly discloses ASC 606 treatment for its Microsoft and NVIDIA multi-year contracts; Crusoe and Lambda are private, and the same service-contract treatment is inferred from their business model rather than from audited filings.

Table A.4: Capital expenditure of additional firms outside the five-firm scope ($ billions)

	2022	2023	2024	2025	Source basis
CoreWeave[†]	—	3.4	9.6	21.5	10-K (cash + OEM-financed)
Crusoe	—	—	∼1.0	∼4.0	private; press anchors
IREN[‡]	—	—	0.5	1.6	10-K (fiscal year ending June 30)
Lambda	—	—	<0.5	∼1.0	private; press anchors
xAI[§§]	—	0.5	5.6	12.7	SpaceX S-1 (AI segment, audited)
Subtotal	**0**	**∼4**	**∼17**	**∼41**	

Notes: "—" indicates that the firm was pre-AI in business model or pre-founding, or that AI-attributable capital expenditure was negligible. [†]CoreWeave's "economic" capex equals cash purchases of property and equipment plus liabilities related to property-and-equipment additions financed by original equipment manufacturers (chiefly NVIDIA). The OEM-financed component grew 12.5× in 2025 as suppliers provided extended-term financing on GPU deliveries (CoreWeave 10-K for fiscal year 2025, filed March 2, 2026, accession 0001769628-26-000010, Consolidated Statements of Cash Flows). [‡]IREN's fiscal year ends June 30; the values reported are calendar-year approximations stitched from quarterly cash-flow disclosures. [§§]xAI was acquired by SpaceX in February 2026 and is reported as the AI segment of the combined entity in SpaceX's S-1, filed May 20, 2026 (Space Exploration Technologies Corp., 2026); the 2023–2025 figures are the audited AI-segment capital-expenditure line in the Consolidated Statements of Cash Flows. Crusoe and Lambda remain private with no audited filings; the values shown for these two firms are best estimates from primary press sources and venture-capital databases, with substantially weaker defensibility than the public-filer values.

proximately $95B, dominated by xAI and CoreWeave) and is expected to continue growing into 2027. For the calibration in Section 3.3, omitting the expanded universe leaves the pre-boom AI capital stock K_{a0} essentially unchanged (the 2022 contribution is negligible) and biases the post-boom investment, and with it the capital stock entering 2028, downward. Because $e^{\hat{\xi}}$ is the ratio of the capital stock entering 2028 to K_{a0}, $\hat{\xi}$ is increasing in post-boom investment, so the five-firm calibration yields a downward-biased $\hat{\xi}$ relative to one that incorporates the expanded universe.

Sources by firm. For **CoreWeave**, FY2023–FY2025 capex is from the FY2025 10-K cash-flow statement (filed March 2, 2026, accession 0001769628-26-000010), with the OEM-financed layer from "Liabilities related to property and equipment additions, including OEM-financed additions" in the same statement. For **IREN**, FY2024 and FY2025 fiscal-year capex (ending June 30) is from the FY2025 10-K (filed August 28, 2025); calendar 2024 and 2025 are stitched from Q3 FY2026 quarterly disclosures (10-Q filed May 8, 2026). For **xAI**, the 2023–2025 figures are the AI-segment line of the Consolidated Statements of Cash Flows in SpaceX's S-1, filed May 20, 2026, accession 0001628280-26-036936 (Space Exploration Technologies Corp., 2026). The S-1 reports the AI segment on a pro-forma basis as if the February 2026 xAI acquisition had occurred at the beginning of the historical period presented, so the 2023–2025 figures correspond to the standalone xAI entity (which also encompasses the X platform following the March 2025 X/xAI merger described in the S-1). Because the S-1's three segments are Space, Connectivity, and AI, the AI-segment capital-expenditure line includes X-platform infrastructure, which the S-1 does not report separately. The S-1 also reports Q1 2026 AI-segment capex of $7,723 million, which is the empirical anchor for the 2026 forecast cited above.[27] **Crusoe** and **Lambda** are private

[27] The same filing reports Q1 2026 capex of $1,052 million for SpaceX's Space segment (launch and Starship development) and $1,332 million for the Connectivity segment (Starlink). Neither figure is included in our AI infrastructure aggregate.

and have not filed audited financial statements. Crusoe's 2024 and 2025 estimates are press-anchored to its revenue trajectory ($276 million in 2024, approximately $998 million in 2025) and its disclosed share of the Stargate Phase 2 joint venture ($15 billion total commitment, with $11.6 billion of debt and equity raised through 2026). Lambda's estimates rely on its November 2025 Series E disclosure ($1.5 billion, bringing cumulative equity raised to over $2.3 billion), with modest pre-2025 spend reflecting an earlier stage of capacity buildout.

Tier 2 partial-overlap firms. Two additional public filers—Applied Digital and Hut 8—are data-center owners with operating-lease customer relationships under ASC 842 rather than ASC 606. When a hyperscaler tenant classifies its lease as a finance lease, the investment may already be reflected in the tenant's reported capital expenditures, so adding the lessor's capex may double-count some investment. Applied Digital's FY2026 Q3 10-Q discloses that the majority of its HPC hosting revenue is recognized under ASC 842 as operating leases; Hut 8's Beacon Point contract is a 15-year triple-net lease that would typically receive finance-lease treatment by the tenant. We therefore exclude Applied Digital and Hut 8 as a conservative measure to avoid double-counting, although this may leave some investment unaccounted for.

B AI-capital depreciation

The baseline rate $\delta_a = 0.25$ captures the relatively rapid depreciation of AI investment, much of which is computing equipment. Under the geometric law of motion (2), 32% of a cohort remains after four years and 18% after six. The non-AI depreciation rate, by contrast, is $\delta_n = 0.07$.

The firms' accounting policies use straight-line depreciation. Between 2022 and 2025, all four—Microsoft, Alphabet, Meta, and Amazon—extended their estimated useful lives

for server and networking equipment: Microsoft and Alphabet to six years (Alphabet Inc., 2023, Microsoft Corporation, 2023), Meta to 5.5 years for certain server and network assets (Meta Platforms, Inc., 2025), and Amazon in stages to six years (Amazon.com, Inc., 2023, 2025). Amazon partially reversed the extension in January 2025, shortening the lives of a subset of its servers and networking equipment from six to five years, citing "an increased pace of technology development, particularly in the area of artificial intelligence and machine learning" (Amazon.com, Inc., 2025). These useful lives do not translate directly into geometric depreciation rates. Straight-line depreciation writes an asset off completely over a fixed horizon, whereas geometric depreciation leaves a fraction of each cohort at every horizon. We consider $\delta_a = 0.15$ as an alternative with more durable capital: 38% of a cohort remains after six years at this rate.

Depreciation affects the inferred boom size through two opposing channels. Recall that (13) identifies the boom size from $K_{a,2028}/K_{a0}$, where the initial stock entering 2023 is $K_{a0} = I_{a,\text{pre}}/\delta_a$. To see the first channel, temporarily set aside depreciation during the investment period, while retaining this initial stock. All investment then adds to the stock, without replacing any capital lost during the buildout. Denoting the resulting hypothetical stock by $\widetilde{K}_{a,2028}$, equation (12) simplifies to

$$\frac{\widetilde{K}_{a,2028}}{K_{a0}} = 1 + \frac{\sum_{t=2023}^{2027} I_{at}}{K_{a0}}.$$

Investment adds dollar amounts to the capital stock. A lower depreciation rate implies a larger initial stock, so the same investment represents a smaller proportional expansion. This channel therefore implies a smaller productivity boom at a lower depreciation rate.

A second effect pushes in the opposite direction. In the actual capital accumulation equation, part of investment replaces depreciation of the initial stock and of capital installed during the buildout. The more investment goes toward replacement, the less remains to

expand the stock, and the smaller the boom needed to justify the observed expenditure. Conversely, at a given initial stock, lower depreciation reduces this replacement requirement and leaves more capital in place in 2028, implying a larger boom.

Restoring depreciation during the buildout and substituting $K_{a0} = I_{a,\text{pre}}/\delta_a$ into (12) gives the exact expression

$$e^{\hat{\xi}} = \frac{K_{a,2028}}{K_{a0}} = (1-\delta_a)^5 + \delta_a \sum_{t=2023}^{2027} (1-\delta_a)^{2027-t} \frac{I_{at}}{I_{a,\text{pre}}}. \tag{23}$$

With the investment data in Table 1, the first effect dominates. Lowering δ_a from 0.25 to 0.15 raises the initial stock from \$636B to \$1,060B, an increase of 67%. The capital stock entering 2028 also rises, from \$2,212B to \$2,755B, but by only 25%. Its ratio to the initial stock therefore falls: $e^{\hat{\xi}}$ is 2.60 rather than 3.48.

Table B.1 compares the baseline (Panel A) with the alternative calibration (Panel B), including the AI capital stock alongside the quantities in Table 7. In each panel, the boom size is inferred from the same investment data using that panel's depreciation rate.

Panel B uses $\delta_a = 0.15$ throughout, with the boom size implied by (23): $e^{\hat{\xi}} = 2.60$, or $\hat{\xi} = 0.96$. Because the calibration of p_* to the price-to-dividend ratio in Section 3.3 identifies expected AI productivity growth $p_*(e^{\hat{\xi}} - 1)$ rather than p_*, Panel B sets $p_* = 0.047$, which holds expected productivity growth at its baseline value of 7.4% per year. Output shares and GDP gains are lower than at baseline in all three scenarios, although in the moderate scenario capital is higher: the post-window stock is \$2,927B against \$2,343B, while output is \$2,530B against \$2,706B. Output per unit of capital in the post-window steady state, $Y_a/K_a^* = (r_a + \delta_a)/(\alpha\,\Omega(p_*))$, is 0.86 in Panel B against 1.16 in Panel A, and the gain in output is measured from a pre-boom level of \$954B rather than \$763B. In the transformative and singularity scenarios, with two and three booms in all, the smaller boom compounds ($e^{2\hat{\xi}}$ is 6.8 rather than 12.1 and $e^{3\hat{\xi}}$ is 17.6 rather than 42.1), and capital is lower than at

Table B.1: Post-window output shares and GDP growth under alternative depreciation calibrations

	AI share (%)	Cum. GDP growth (%)	AI capital ($B)	Post-window I_a ($B/yr)
Panel A. Baseline: $\delta_a = 0.25$, $e^{\hat{\xi}} = 3.48$, $p_* = 0.03$				
Pre-boom	3.2	—	636	159
Moderate	10.4	8.1	2,343	586
Transformative	28.7	35.9	8,149	2,037
Singularity	58.4	132.6	28,347	7,087
Panel B. $\delta_a = 0.15$, *re-estimated* $e^{\hat{\xi}} = 2.60$, $p_* = 0.047$				
Pre-boom	3.9	—	1,060	159
Moderate	9.8	6.5	2,927	439
Transformative	22.0	23.1	7,608	1,141
Singularity	42.3	66.4	19,770	2,966

Notes: Construction parallels Table 7. "AI capital" is the pre-boom stock $K_{a0} = I_{a,\text{pre}}/\delta_a$ in the pre-boom row and the post-window steady-state stock K_a^* otherwise; post-window $I_a = \delta_a K_a^*$. Cumulative GDP growth is the one-time increase in the level of GDP from the pre-boom economy to the post-window steady state. Pre-boom investment is the 2022 capital expenditure of $159B; non-AI output, r_a, r_n, and δ_n are at their baseline values in both panels. Panel B sets p_* so that expected AI productivity growth $p_*(e^{\hat{\xi}} - 1)$ equals its baseline value of 7.4% per year. Calculations use the unrounded values of $\hat{\xi}$ and, in Panel B, $p_* = 0.0465$.

baseline as well. With booms arriving from 2031 on at the rate p_* of each panel, as in Figure 3, expected cumulative GDP growth by 2050 is 36.5%, 101%, and 269% in the three scenarios, against 40%, 148%, and 522% at baseline. The recalibration holds expected AI productivity growth after 2030 fixed at its baseline value. The lower GDP gains reflect the smaller productivity gains during the boom window, with capital adjustment also affecting the subsequent output path.

C Frictionless investment paths

Figure C.1 shows the investment paths retaining the annual investment data through 2027, then adjusting immediately to K^* from the 2028 decision onward. The capital stock entering 2028 equals $K^*(z_1, 0)$, as in the partial-adjustment exercise, so the 2028 investment of \$3,021B is the frictionless response to the rise in the boom probability. Frictionless adjustment produces sharp investment changes and can require negative investment when p reverts to p_* without a new boom. Compare with the smoothed paths in Figure 1.

Figure C.2 is the counterpart of Figure 3 with capital at its optimal level $K^*(z_{at}, p_*)$ each year from 2030 on, in place of the partial-adjustment rule. The 2030 values are then the fully adjusted outcomes of Table 7, which produces the step at 2030 in the singularity scenario, and expected growth by 2050 is 44%, 161%, and 567% in the three scenarios.

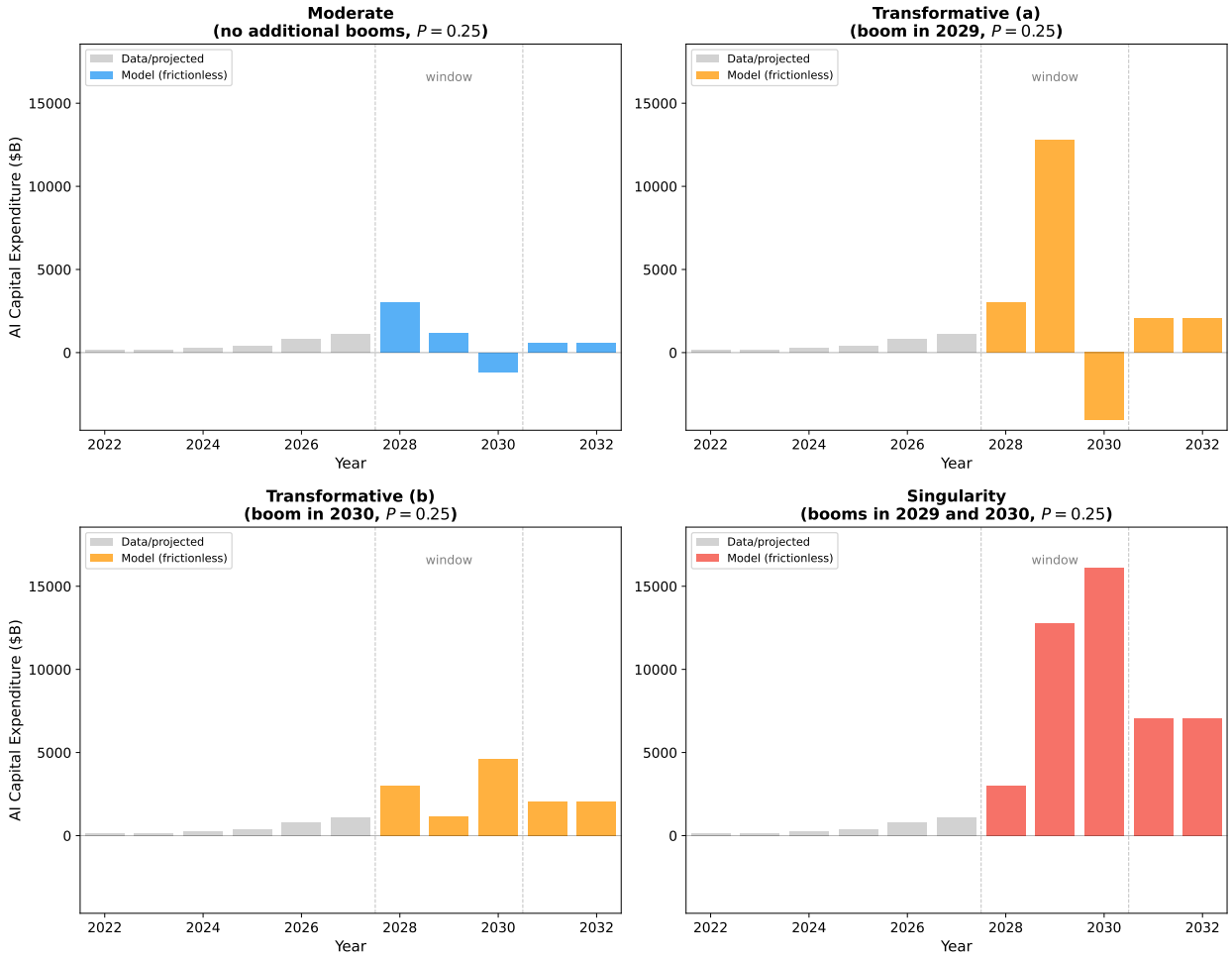


Figure C.1: Frictionless investment paths by scenario. Gray bars show actual and projected AI capital expenditure through 2027 (Table 1). Colored bars from 2028 onward show frictionless investment, starting from the capital stock entering 2028, which equals the calibrated target $K^*(z_1, 0)$. Dashed lines delineate the high-probability window (2028–2030).

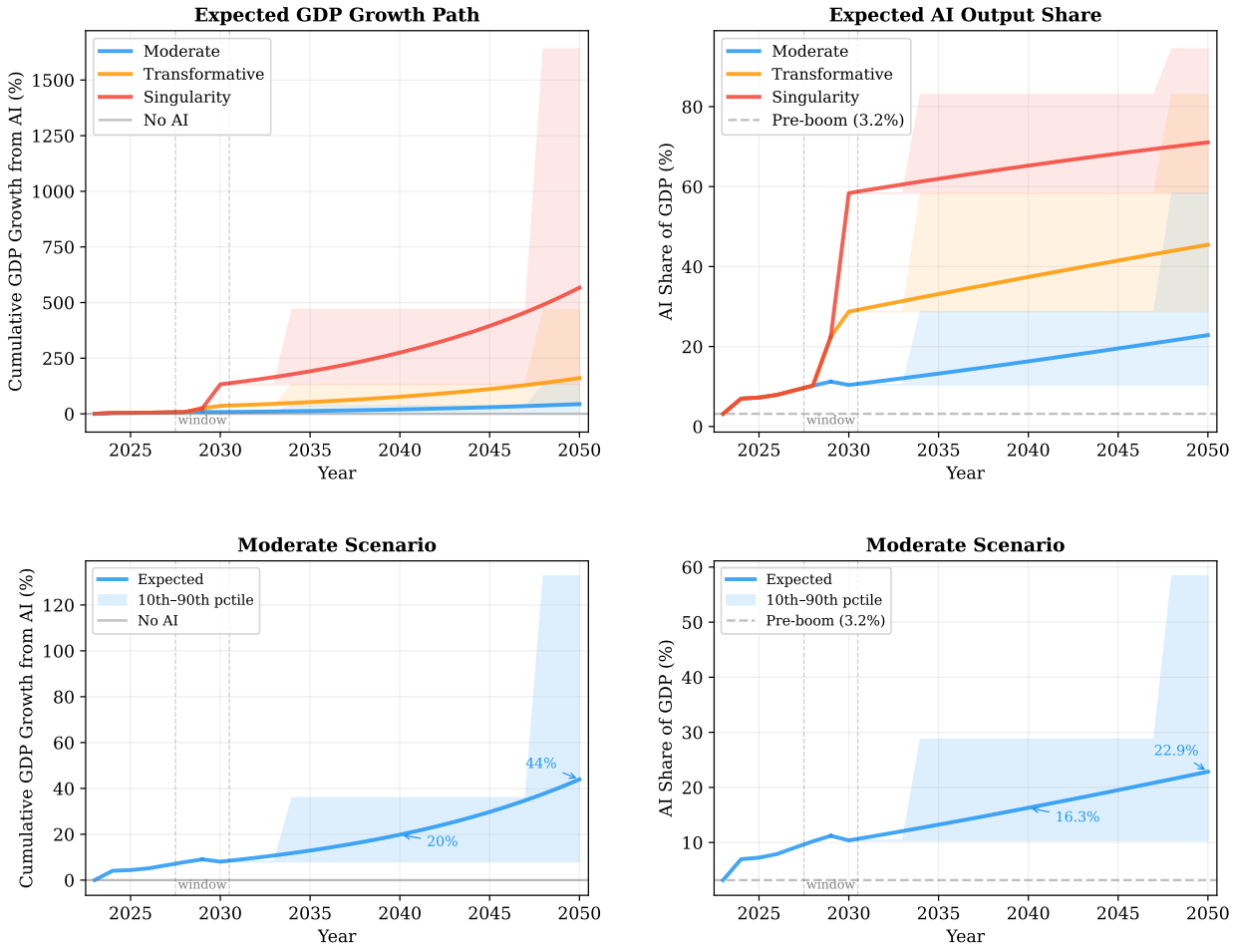


Figure C.2: GDP growth distributions (2024–2050) with capital at its optimal level from 2030. As Figure 3, except that from 2030 on capital is set to $K^*(z_{at}, p_*)$ each year; the years 2024–2029 use the partial-adjustment paths.